

English (stative) passive participles: Category, argument introduction, and interpretation*

Alison Biggs and David Embick

McMaster University and University of Pennsylvania

ABSTRACT We develop a syntactic analysis of *stative passive participles* in English: cf. *the flattened box...*, *The box is flattened*. One focus is on new generalizations about stative passive interpretation: in predicative position the participle has a stative interpretation, while in attributive position it has both a stative and a purely eventive interpretation; this leads to the question of why these meanings are distributionally conditioned. A second focus concerns the stative passive's eventive interpretation. This raises the question of how stative passive participles relate to the *eventive passive participle*, seen in *The box was flattened by Susan*. The two participles share a non-verbal distribution; they are also invariably identical in form in English. Yet they differ as well: the eventive passive has only an eventive interpretation, while the stative participle cannot be eventive in predicative syntax. We explain the similarities between the two passives by positing a head α in both. This head defines a non-verbal syntactic distribution, and accounts for the shared form. For the differences, what α attaches to is different in the two participles: in the eventive passive, the complement of α is a phrasal VoiceP; in the stative passive, it is a *small* (non-phrasal) verbalized structure (i.e. a complex head). We account for the syntactically-sensitive interpretations of the stative passive in connection with *argument introduction*: when there is an argument introduced by the head i^* , the interpretation of α is stative; otherwise the interpretation is eventive. This (*contextual*) *allosemy* of α provides an explanation for the attributive/predicative asymmetry by admitting an eventive interpretation only when the participle does not introduce its own argument, as is possible in attributive syntax. Taken as a whole, this analysis requires an *abstract* view of syntactic categories, in which a particular category like α does not necessarily correlate with a single type of argument introduction, or a single interpretation (stative or eventive). Our discussion examines the implications of this view, and questions concerning the relation between α and the category 'adjective', the nature of allosemy, and the prospects for exploring these matters cross-linguistically.

KEYWORDS: stative passive, category, participles, allosemy, Distributed Morphology, event structure

1 Introduction

The nature of syntactic categories is a central question in grammatical theory. One of the most important questions that arises in this domain concerns how category-membership does or does not correlate with other properties: in particular, with particular interpretations, or with particular ways of introducing arguments. A commonly-held view is that such relations are direct. Membership in a particular syntactic category thus carries certain expectations; for example, it might be held (to a first approximation) that nouns are interpreted as entities, that verbs are interpreted as events, that adjectives are states, and so on; and that each of these categories introduces arguments (or does not do so) in ways that are specific to particular categories.

Cross-categorical effects are particularly relevant to this line of inquiry. If e.g. nouns derived from verbs have 'mixed' properties, then they promise to reveal much about both the syntax of different categories, and how they relate to semantic interpretation. Adding argument introduction into the picture offers a further interaction to be explored. In this case, the question is whether deverbal structures introduce arguments in the same way that verbs do; more generally, whether the distribution of nominal

*Thanks to...

arguments behaves in a single way across categories, or exhibits differences. All of these questions (and more) are investigated in Chomsky (1970), much of which discusses the consequences of dispensing with the distinction between lexical categories and grammatical features that was part of earlier generative approaches. Chomsky highlights the possibility that typical lexical category labels ('Noun', 'Verb', 'Adjective') might be too coarse; that is "...the reflection of a deeper feature structure, each being a combination of features of a more abstract sort." He does not explore this position in depth, however, regarding it as "...hardly clear enough to be a speculation" (1970:199).

One way of approaching the kind of abstraction that might be at play here is found in theories that employ *abstract morphemes*. Frameworks like Distributed Morphology and related approaches have devoted a great deal of attention to how such morphemes, which are viewed as purely syntactic objects, receive their forms and semantic interpretations at the PF and LF interfaces. Looking at the questions raised by category in particular, categories in this theory are defined by categorizing heads that are combined with category-neutral Roots. Treating the categorizers as abstract morphemes raises the possibility that associations between category, interpretation, and argument introduction might be somewhat *indirect*. How a particular category-defining head *x* behaves is in principle a function both of its inherent feature content, and of the syntactic context in which it is interpreted at the PF and LF interfaces. We thus might expect to find members of the same lexical category interpreted differently, or behaving differently with respect to arguments, in ways that are determined by their local syntactic context.

This paper develops an argument for an abstract view of category of the type just described. Our specific focus is on participles, which are traditionally taken to possess properties of both verbs and adjectives. We begin with an examination of what we will refer to as the *stative passive participle*, illustrated in the examples in (1):

- (1) a. The **flattened** boxes ...
- b. The boxes are **flattened**.

These are (perhaps more commonly) referred to as *adjectival passive participles* in the literature; we will avoid this term since it presumes answers to questions that are controversial.

To a first approximation, stative passive participles are deverbal forms—i.e., they are built from Roots that typically appear as verbs— but they have a syntactic distribution typical of modifiers. Compare (1) with the typical adjectives in (2):

- (2) The **flat** boxes.../The boxes are **flat**.

The stative passive participle is different in important ways from the eventive passive participle in (3). For example, though the stative passive in (1b) is string identical to the first part of (3), the interpretations are distinct. With (1b) the assertion is that the boxes are in a flattened state at a particular point in time; (3), however, is habitual, in a way that is typical for English eventive clauses in the present tense: it describes an event that happens every day in a particular way.

- (3) The boxes are **flattened** with a hammer every day at noon by the toddlers.

Although the examples in (1)-(3) differ in meaning and in certain syntactic properties, the participles crucially share a number of properties, both in terms of form (where there is identity) and in terms of syntactic distribution. To account for these similarities, many prior analyses have gone so far as to derive the stative passive from the eventive passive; however, these accounts are untenable for reasons that will become clear in the pages to come.

One of the primary empirical points established in this paper concerns the interpretation of stative passive participles. We will show that this participle has two interpretations: one stative (as the name

implies), but also a purely eventive one. Crucially, we will show that the eventive interpretation is available only when the stative passive participle is in attributive syntax. This novel observation adds a new explicandum to the question of what the stative and eventive participles do and do not share. There are puzzles on the syntactic side as well. For example, the internal structure of the eventive passive behaves differently from that of stative passives with respect to phrasal modifiers, again in ways that will be made precise below.

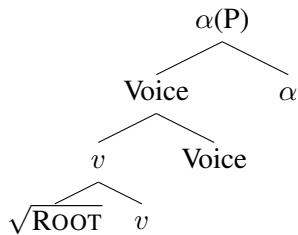
Bringing these different threads together, at least the following observations require explanation:

- **Internal syntax:** The eventive passive participle and the stative passive participle differ with respect to phrasal modification.
- **Form and distribution:** The eventive passive participle and stative passive participle share formal and distributional properties.
- **Meaning1:** The stative passive can have a stative interpretation in both attributive and predicative positions.
- **Meaning2:** The eventive passive is interpreted eventively in both attributive and predicative syntax. The stative passive, on the other hand, has an eventive interpretation only in attributive syntax.

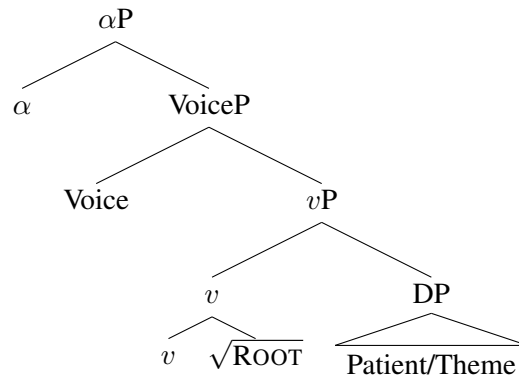
The analysis we develop is centered on two core proposals. We will briefly introduce these here, leaving full discussion for the main body of the paper where the details will be motivated in context.

Small versus phrasal Building on recent work, we will demonstrate that the stative passive has a *small* syntactic structure, in which there is no phrasal material in the complement of the participle-defining head. Simplifying for expository purposes, this participle consists of a verbalized Root, along with a head given here as α for reasons we will explain immediately below, as shown in (4). By way of contrast, the eventive passive participle has α taking a phrasal complement, as shown in (5):

(4) stative passive (small)



(5) eventive passive (phrasal)



We assume that affixation takes place in the eventive passive to produce a complex head that is identical to (4). The identity in participle form follows from this structural identity. Another property shared by the two types of participles also derives directly from the presence of α in both structures: this head defines the same **external** syntactic distribution (=appearance in both attributive and predicative positions). The **internal** syntactic differences between the two participles with respect to phrasal modification follow from the small versus phrasal distinction: eventive passives have phrasal structure embedded under α where XP modifiers may attach, while stative passives do not.

An abstract approach to category The way in which the head α functions in our analysis makes use of the idea that morphemes are abstract; a central component of Distributed Morphology and other theories with *Late Insertion* at the PF and LF interfaces. Abstract morphemes provide a way to analyze participles such that they share the head α that defines their distributional properties, as just described. At the same time, α is interpreted at the PF and LF interfaces in ways that depend on the syntactic context in which it appears. On the PF side, α displays allomorphy that is conditioned by the Root, and has the same form in both participles. On the meaning side, we argue that how α is interpreted also depends on its immediate syntactic context. When it appears by itself, as in the structures in (4) and (5), α has no semantic interpretation; the result is an eventive interpretation. However, when α is the complement of an argument-introducing head, it receives a stative interpretation. This (*contextual*) *allosemy* of α allows us to explain both why stative passives may have both eventive and stative interpretations. It also provides a basis for the fact that these interpretations are associated with the syntactic position (attributive vs. predicative) in which the participle appears, as we will explain in the main text.

In short, the small versus phrasal distinction and the abstract approach to category together provide an explanation of the properties shared by the two passive participles, as well as how they differ.

The body of the paper proceeds in the following sequence:

- First, §2 clarifies the empirical ground concerning how stative passives are interpreted in English, elaborating on a prior literature that is sometimes inconsistent or unclear on key points. The primary novel observation is that while (as has been proposed in the literature) stative passive participles may have both stative and eventive interpretations, the latter is restricted to attributive syntax.
- In §3 we motivate the small analysis of stative passive participles. We show that a phrasal analysis of stative passives makes incorrect predictions about modification patterns. The correct predictions are produced by an analysis in which XP modifiers of stative passives are introduced only external to the participle-creating head.
- In §4 we articulate an approach to category and argument introduction, and show how it explains the attributive/predicative asymmetry: in particular, why the eventive interpretation of the stative passive is restricted to attributive syntax.
- §5 further examines the distribution of eventive interpretations in participles in detail. If eventive passive participles are able to attach as attributive modifiers, then it is in principle possible that all eventive interpretations are produced by eventive participles, rather than by stative passive participles. Contrary to this hypothetical, we introduce diagnostics that confirm that stative passive participles are indeed interpreted eventively.
- The primary theoretical results of the paper are reviewed in §6, which examines (i) the implications of the approach for the theory of category— with a focus on α and the category *adjective*; and (ii) the nature of allosemy, and what kinds of cross-linguistic predictions are made by analyses that do and do not appeal to meanings determined by local structural context.
- §7 offers a general conclusion.

Before proceeding, there is a terminological point that deserves special emphasis. We will use the term *stative passive participle* to refer to the particular syntactic object in (4), irrespective of its meaning. In addition, for both participle types, we will sometimes omit the word ‘participle’ and use *stative passive* or *eventive passive*.

2 Clearing the empirical ground

This paper is centered on the analysis of stative passive participles like that in (6a); for the reasons introduced above, such an analysis must also take into account similarities to and contrasts with simple adjectives (6b) and eventive passives (6c).

- | | | | |
|-----|----|--|--------------------------------------|
| (6) | a. | The tent is <i>flattened</i> . | <i>stative participle participle</i> |
| | b. | The tent is <i>flat</i> (on the ground). | <i>adjective/‘pure stative’</i> |
| | c. | The tent is <i>flattened by the wind</i> at least once per season. | <i>eventive passive participle</i> |

The first question to be addressed concerns how different types of eventualities— i.e., events and states— figure in the interpretations of (6a-c). It is generally agreed that adjectives like *flat* in (6b) are purely stative (hence the alternate way of referring to them). As typically described, stative passive participles as in (6a) are deverbal and stative, and are in addition associated with an event in a way that simple adjectives are not. The stativity in (6b) contrasts with the corresponding eventive passive participle in (6c), which is eventive in the same way as its corresponding active (and hence habitual in the present tense).

This section revises this initial characterization by establishing that stative passive participles in fact have two distinct interpretations. In one, there is a state that is the end of an event; in the other, the interpretation is purely eventive. The latter interpretation has not been clearly identified in prior work on the stative passive, and so establishing its availability is our first goal in this section. The second goal is to show that the two interpretations are distributionally conditioned: the stative interpretation is found with stative participles in both attributive and predicative syntax, while the eventive interpretation is available only when the participle is in attributive position.

2.1 States and events

Simple adjectives are typically taken as property ascriptions of their subject’s referent. There is a general consensus that stative participles differ semantically from simple adjectives in ascribing a complex property, with the verbal base contributing an event (e.g. Levin and Rappaport, 1986; Kratzer, 2001, and much related work), in addition to the stative component of their interpretation. The presence of the event means that the stative participle can be employed to assert that a state is produced by an event, whereas the pure state is neutral with respect to any event having brought it about; this is the contrast between (6a) and (6b).

Some simple examples suffice to further illustrate the contrast. For example, in a scenario in which a number of boxes are lying flat on the ground, it is possible to assert either of the sentences in (7a-b). However, stative passive (7a) requires that the boxes be flat as a result of an event of flattening, while the adjective (7b) does not. Thus, the adjective is compatible with the assertion that the boxes were manufactured in the flat state (8b), while the participle is not (8a) (Embick, 2004):

- | | | |
|-----|----|----------------------------------|
| (7) | a. | The boxes are <i>flattened</i> . |
| | b. | The boxes are <i>flat</i> . |
| (8) | a. | *The boxes were made flattened. |
| | b. | The boxes were made flat. |

Relatedly, pure adjectives can be Resultative Secondary Predicates, while stative passive participles cannot, as shown in (9) (Embick, 2004, 2023):

- (9) Mary kicked the boxes {open,*opened}; {flat;*flattened}

While previous accounts have not reached a consensus concerning precisely how the stative and eventive components relate to each other, the idea that the stative interpretation of the participle involves an event that stands in some nontrivial relation to a state is uncontroversial.

The position we will develop is that English stative passive participles have two distinct interpretations, given in (10):

(10) *Interpretations*

- a. @-interpretation: Argument is in target state of Root-identified event at Topic Time.
 $\lambda x \lambda s \exists e \sqrt{\text{ROOT}}(e) \wedge \text{End}(e, s) \wedge \text{Theme}(x, s)$
- b. E-interpretation: Argument underwent Root-identified event prior to Topic Time.
 $\lambda x \lambda t \exists e \sqrt{\text{ROOT}}(e) \wedge \text{Patient}(\text{Theme})(e, x) \wedge \tau(e) < t$

The @ interpretation First, there is what we will call an @-*interpretation* as in (10a), in which a Theme is the holder of a state that is related to an event; for this use of Theme, see e.g. Parsons (1990) and the discussion in Williams (2015). This is the interpretation described above in relation to (7) and (8). We employ a relation End in (10a) that relates an event e to the state s that the event terminates in; cf. Williams (2015) on change of state verbs, building on Pietroski (2005). For the @-interpretation, the crucial point is that the state holds **at** the moment of evaluation, which we refer to as the Topic Time (Klein, 1994).¹

The E-interpretation Second, there is an *E(ventive)-interpretation*, which specifies that an argument underwent an event of the relevant type at an anterior time, as in (10b).² The crucial point is that the E-interpretation asserts simply that an event took place at an anterior point, with the argument in question being thematically related to that event; it is completely neutral with respect to whether or not a corresponding state obtains at Topic Time. The E-interpretation is essentially the same as that found with the eventive passive participle. Thus, in §4-§5 we develop specific proposals that explain what these participles do and do not share in terms of structure, distribution, form, and meaning.

Before we proceed, a few comments are in order concerning how the primary claims of this section relate to prior literature. The idea that stative passive participles might be associated with more than one meaning has been a central point of study since at least Kratzer’s (2001) influential study. Kratzer analyzes German stative participles as having either a *Resultant State* or a *Target State* reading, a distinction adapted from Parsons (1990). In important respects, our E- and @- interpretations are similar to the Resultant and Target state interpretations, respectively.

One of the critical ways in which the approach in (10a-b) departs from previous work is our claim that the availability of the two interpretations is distributionally conditioned. In particular, the interpretation of stative participles in English is sensitive to attributive versus predicative syntax. Stative participles in predicative syntax have only a stative interpretation. In contrast, we show that stative participles in attributive syntax can be interpreted either statively and eventively. We develop this idea in §2.2 and §2.3, and summarize the generalizations to be accounted for in §2.4.

¹While for immediate purposes the single event in (10a) will suffice, we believe that there are reasons for treating the end-state as the end of an abstract event of causation that is distinct from a Manner event; this brings the analysis closer to the use of End in the analysis of change-of-state verbs in the works cited immediately above. However, the simple semantics in (10a) is sufficient for the purposes of this paper. For that matter, for much of what we have to say it would be possible to employ a Cause relation relating an event to a state, as in Kratzer (2001) and related work.

²In (10b) we give the role as Patient(Theme), to indicate that the role is whatever is assigned by the corresponding verb. For example, there would be a Patient for, e.g., *kick*, but a Theme for *flatten* etc.

2.2 Interpretations and truth conditions

The two interpretations in (10)– and their distributional availability– can be identified by determining whether or not the truth of a participle-containing utterance depends on a type of state holding. To start with, consider the predicative participle in (11).

(11) The box is flattened.

This sentence has an @-interpretation, which specifies that the box in question is *in a flattened state*: the box holds the state in question at the moment of evaluation. This interpretation is straightforward.

In contrast, an E-interpretation would require only that the boxes in (11) *have undergone a flattening event before*; an E-interpretation does not specify that the state holds at Topic Time. The interpretive difference between the @- and E- readings can be brought out in context. The scenario in (12) prompts an E-reading: it requires that the box have undergone a flattening event but does not require that a resulting state holds at Topic Time. The stative participle in predicative syntax in (12) cannot be interpreted in this way; which is to say, the E-interpretation is not available. Instead, (12) only has the @-reading.

(12) *Scenario: There are two cardboard boxes on a table. Carrie flattens one. Then she puts it back into its assembled form. <Pointing to the one that underwent the flattening>:*

*This box is flattened.

*E-interpretation

The judgment in (12) is unexpected for analyses that are based on Kratzer (2001), whose analysis of German says that (something like) an E-interpretation should be freely available. However, (12) (and its context) show that the E-reading is not available for predicative participles, at least for English. The absence of the relevant reading in examples like (12) has in fact been often previously observed, for both English and German (see e.g. Maienborn, 2011; Gese, 2011; Gehrke, 2015; Maienborn et al., 2016; Ramchand, 2018).

Crucially, the E-interpretation that is absent in predicative (12) **is** available when the participle is in attributive position. Using an identical context to that in (12), the E-reading for an attributive participle is illustrated in (13):

(13) *Scenario: There are two cardboard boxes on a table. Carrie flattens one. Then she puts it back into its assembled form.*

The (recently/previously) flattened box (... is on the left)

The example in (13) demonstrates that it is perfectly natural in English to use the stative participle to describe a box that has undergone a flattening event, even in a context in which the flattened state does not hold at Topic Time. The adverbs *recently*, *previously* are employed in (13) to facilitate the E-interpretation. Crucially, the adverbs that facilitate the E-interpretation in (13) do not help the participle in predicative position (14).

(14) #This box is recently/previously flattened.

To the extent that (14) is felicitous, it has an @-interpretation that requires the box to be in the flattened state at Topic Time, with the fact that the flattening was *recent* or *previous* somehow manifesting itself in that state. It is sometimes possible to coerce such readings, given appropriate context. For this reason, we have marked (14) with a hashtag #. We will have more to say about this effect in §3, where we examine the modification of participles in detail.

The E-interpretation of attributive participles is not the only possible one; an @-interpretation is available in attributive position as well, illustrated in (15):

- (15) *Scenario: There are two boxes on a table; the one on the left is flat, the one on the right is in a typical box shape.*

The flattened box (...is on the left).

@-interpretation

2.3 Root distribution

The distributional aspect of stative participle interpretation is also clear with respect to how different types of Roots behave. Root-specific factors play a key role in acceptability judgments of stative passives. Perhaps the most commonly made observation concerns whether a Root is associated with a typical end state or not (Kratzer, 2001; Anagnostopoulou, 2003; Alexiadou et al., 2015; Beavers and Koontz-Garboden, 2020, among others). To a first approximation, it is generally agreed that Roots that are associated with accomplishment interpretations as in (16a) are much more felicitous as stative passives than those that lack an end state, as in (16b). Note that the participles are in predicative position.³

- (16) a. These boxes are flattened/opened/emptied/crushed/destroyed ...
b. #These boxes are kicked/touched/poked/examined ...

Other diagnostics produce robust correlations with the contrasts in (16). One example is durative *for-X-time* modification. This modifier has often been employed to show that e.g. *flatten* has an end state, while e.g. *kick* does not, as illustrated with active transitive verb phrases in (17)-(18):⁴

- (17) They *flattened* the boxes for an hour.
a. Event modification: One hour of box-flattening events.
b. State modification: Maintained the boxes in a flattened state for one hour.
(18) They *kicked* the boxes for an hour.
a. Event modification: One hour of box-kicking events.
b. State modification: *

The contrasts illustrated above, along with others that will be introduced as discussion proceeds, have led to a characterization of Root distribution in Stative Passives along the following lines:

- (19) Two types of Roots
a. Roots associated with end states are felicitous as stative passives.
b. Roots not associated with end states are typically less felicitous as stative passives.⁵

³Although there is general agreement on the facts in (16), prior analyses interpret the effect in different ways. Kratzer (2001) builds the contrast into the semantics, restricting Target States to verbs that introduce a Target State in their lexico-semantic denotation. Other accounts of German are significantly more flexible; Maienborn (2009) argues that, with contextual help, the reading is available for almost any verb; see also Gese (2011); Gese and Hohaus (2012); Gese (2013). For relevant discussion in relation to English, see Embick (2004); Ramchand (2018); Baglini and Kennedy (2019); Beavers and Koontz-Garboden (2020).

⁴Cf. Kratzer (2001). Dowty (1979) develops *for-* modification as an aspectual diagnostic; see also Tenny (1987); Krifka (1989) for different approaches to its use.

⁵The generalization in (19b) is stated weakly (in terms of felicity and not grammaticality) because there are circumstances under which the Roots such as $\sqrt{\text{KICK}}$ are felicitous as statives (cf. Kratzer, 2001; Anagnostopoulou, 2003; Embick, 2004; Maienborn, 2009; Alexiadou et al., 2015; Ramchand, 2018). The felicitous reading is available in certain contexts, sometimes called a 'Job is Done' scenario, as in example (i) from Embick (2004):

- (i) SCENARIO: As employees in a tire factory, we must kick each tire once before quitting time in order to assure quality control.

Although there is some consensus concerning (19), we observe it requires revision once the distributional asymmetries identified in the previous section are considered. To the extent that (19) is correct, it describes Root behavior only in predicative stative passives. In Attributive position, the Root-related contrasts disappear, as shown in (20):

- (20) a. The (recently) flattened/opened/emptied/crushed/destroyed box...
 b. The (recently) kicked/touched/poked/examined box...

The contrasts between (16) and (20) can now be framed in terms of the generalizations that we have developed in this Section. In predicative position only an @-interpretation is available, and the Roots that lack an end state in (16b) are consequently odd. In contrast, in attributive position there are both @- and E-interpretations. Since the E-interpretation should be freely available for any Root (not just Roots associated with end states), the acceptability of attributive (20b), in contrast to predicative (16b), is expected.

2.4 Summary

We have identified a simple contrast in the interpretation of stative participles in attributive and predicative syntactic positions, summarized in (21):

- (21) *Stative passive interpretations*
 a. Attributive position: Both @- and E-interpretations
 b. Predicative position: Only @-interpretation

Throughout the remainder of the paper we provide further evidence that reinforces the observation that the E-interpretation is available only in attributive syntax.

To this point, the E-interpretation has been shown to be narrowly restricted to (prenominal) attributive syntax. Of course, stative passive participles appear in more contexts than just the predicative and attributive seen thus far. These additional environments include complements of main verbs such as *remain* and *seem* (and others), as well as depictive secondary predicates (22):

- (22) a. The box remained **flattened** for three hours. @
 b. The box seemed **flattened**, but in reality... @
 c. The box arrived **flattened**. @

In these further contexts, it seems that only the @-interpretation is found. For example, (22c) is not true if the box arrived in fully assembled form, but had been flattened at some point in the past. As the structural contexts in (22) introduce additional complexities, we employ examples with participles under predicative *be* in the pages to come.⁶

Ok, *the tires are kicked*, let's go home.

The idea here is that the context provides a relevant sense for *kicked* to be interpreted as an (end) state (thus allowing the task item to be checked off of a list); for another effect that might be related to this one, see also Welke (2007).

Based on the diagnostics that we employ in this paper, it can be concluded that the Job-is-Done interpretation is in fact stative, see Biggs and Embick (2025) for specifics.

⁶To illustrate one of the relevant complexities, the complement of *remain* seems to be restricted to specific types of state interpretation, viz. where the state is potentially transitory. Thus, *The box remained flattened* is felicitous, since a flattened box can be returned to another state (built); but cp. *#The tomato remained flattened*, since tomatoes cannot typically return from such a fate. Since the restriction to transitory states has direct implications for the judgments to be tested, we consequently exclude it from our main discussion except when it serves a specific purpose (one such application appears in §5).

3 The structure of stative passives

Our analysis of the stative passive proceeds in two steps. First, in this section, we develop a syntactic analysis of the ‘core’ structural properties of this participle. With this structure in hand, §4 then shows how this syntactic analysis— in combination with a particular view on how arguments are introduced— derives the @- and E-interpretations, the attributive/predicative asymmetry, and further facts concerning the eventive passive participle.

The prior literature can, as a first step, be divided into two markedly different approaches to the analysis of stative passives: one holds that stative passive participles have no internal syntactic structure, while the other holds that stative passive participles have the same internal structure as eventive passive participles, or at least a great deal of typical verb phrase structure. For ease of reference, we refer to these as *lexical* and *phrasal* analyses, respectively. Beyond (in some sense in between) these two types of analysis is a third approach – to be adopted here – according to which the stative passive participle is built syntactically, but *small*: with a verbalizer *v* (and perhaps a Voice head), but crucially without the internal structure of a phrasal *vP* (Embick, 2023).

One of the most important ways of distinguishing the predictions of these approaches is by comparing how modifiers behave with stative passives on the one hand versus actives and eventive passives on the other. One line in the previous literature identifies something of a puzzle on this point. While it is generally agreed that the stative passive involves an event, it has been frequently observed that this event appears to differ sharply from the simple event assertions that are available with the eventive passive, as diagnosed by various kinds of event modifiers (Rapp, 1996; von Stechow, 1998; Maienborn, 2007; Gese, 2011; Gehrke, 2012, 2015; McIntyre, 2013, 2015; Alexiadou et al., 2015; Maienborn et al., 2016; Ramchand, 2018). Put simply, it appears that the event in the stative passive is not accessible to modification in the way that it is accessible in eventive passives or actives.

Extending the arguments presented in Embick (2023), we will show in §3.1 why modification patterns in stative passives behaves exactly as predicted if stative passive participles have a small syntactic structure, and extend this analysis to explain a further set of facts in §3.2. In §3.3 we introduce the generalizations developed in §2 into the picture, showing that the distribution and interpretation of adverbial modifiers further reinforce the conclusions we reached concerning the attributive/predicative contrast for interpretation.

3.1 A ‘small’ syntactic analysis

As noted above, most prior analyses of stative passives fall between the extremes represented by *lexical* analyses (=no internal syntactic structure) and *phrasal* syntactic analyses (a.k.a. “phrasal layering”, in the sense of Wood (2023)).

Beginning with lexicalist approaches, an important thread growing out of early work on the topic, associated with Wasow (1977) in particular, is that adjectival passives (corresponding roughly to our stative passives) are created in the lexicon (see also Freidin (1975) and Bresnan (1978)), whereas verbal passives (our eventive passives) are formed transformationally. A lexical view is developed (in different forms) in much subsequent work, including, but not limited to, Bresnan (1982); Levin and Rappaport (1986); Dubinsky and Simango (1996); Horvath and Siloni (2008); Meltzer-Asscher (2011); Horvath and Siloni (2019).

The central predictions of the lexicalist analysis are based on the idea that stative participles do not have internal syntactic structure. Instead, stative participles are adjectives created from verbs in the Lexicon, and enter the syntax as a terminal node. They are predicted not to take phrasal *vP* modifiers, as they do not contain a phrasal *vP*. The process that creates these participles could take different forms; one is a rule that creates an adjectival participle from an eventive passive participle, as given schematically

in (23); cf. Levin and Rappaport (1986):⁷

(23) eventive passive participle $\xrightarrow{\text{lexical rule}}$ adjectival passive participle.

The idea that stative participles enter the syntax as terminals, which is to say, with no internal syntactic structure does not mean that stative participles cannot be modified (see especially Meltzer-Asscher, 2011). Instead, the prediction is that *all else equal* stative participles should be modified in the same ways as other members of their syntactic category, i.e. adjectives.

The *all else equal* qualifier has some important details attached to it. One question to ask is whether lexically formed stative passives are expected to accept **all and only** those modifiers that are found with typical adjectives, or whether there might be differences. Empirically, the question is whether a ‘deverbal adjective’, like, e.g., stative passive *flattened* (vs. *flat*) needs to be specified for some feature(s) that distinguish it from a pure adjective. As it turns out, distinctions of this sort are required to explain basic distributional asymmetries; see, for example, discussion around (9). There are different ways of accounting for this kind of effect in a lexical approach; see e.g. Meltzer-Asscher (2011).

The kind of response to participle/adjective contrasts that is most directly relevant to our purposes is found in work by Kratzer (2001) and others, who have emphasized adverbial modification asymmetries between stative passives and adjectives of the type shown in (24):

(24) This box is carefully opened/*open.

Kratzer (2001) argues that the explanation for the contrast in (24) is structural: the stative passive contains a phrasal *vP* to which the AdvP can attach, while regular adjectives lack internal structure. The inability of the manner XP adverb to attach to the adjective follows from this structural contrast. On approaches like this, stative passives have internal phrasal structure; at the least, a phrasal *vP*. Many accounts have adopted this position, including Anagnostopoulou (2003); Embick (2004); Alexiadou and Anagnostopoulou (2008); Sleeman (2009, 2011); Bruening (2014); Alexiadou et al. (2015); Ramchand (2018); and Bešlin (2023).⁸

Although widely adopted, the idea that phrasal structure is required in stative passive participles is not uncontroversial. Concentrating on English, Embick (2023) reexamines the question of putative phrasal structure in detail. A starting point is the observation that if stative passives contain a phrasal *vP*, then (all else equal) stative passives should allow exactly the same kinds of modifiers that are found in corresponding eventive passives. However, this is not the case. Many modifiers that are perfectly felicitous in eventive passives are deviant with the stative passive (Rapp, 1996; von Stechow, 1998; Maienborn, 2009; Meltzer-Asscher, 2011; McIntyre, 2013; Gehrke, 2015).⁹

(25) a. The city was destroyed { rapidly
2035 years ago
with fire arrows
by the Romans

⁷The analysis in (23) assumes that eventive passive participles are created lexically as well. It is not necessary to assume this on a lexical account. However, as discussed by Levin and Rappaport, this derivation provides an account as to why the two participles share a number of properties.

⁸The characterization in the text leaves open the possibility that other projections found in typical verbal clauses like Voice, or certain types of Aspect, might be present in the participle too. We will sometimes use ‘*vP* etc.’ to take these additional heads into account.

⁹The examples in (25b) are annotated with ‘#’ to foreshadow an important point to be addressed in §3.2: modifiers of stative passive participles can be sometimes coerced so that they are much more acceptable.

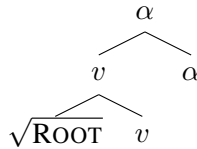
- b. #The city is destroyed $\left\{ \begin{array}{l} \text{rapidly} \\ \text{2035 years ago} \\ \text{with fire arrows} \\ \text{by the Romans} \end{array} \right.$

The question at issue is why there should be any contrast between the (25a) and (25b) examples at all: if both eventive and stative passives have phrasal *v*P (etc.) projections below the participial head, any contrast in modification requires explanation (Maienborn, 2009; Meltzer-Asscher, 2011; Gese, 2011; Alexiadou et al., 2014, 2015; Gehrke, 2015; Maienborn et al., 2016; Ramchand, 2018). One possibility is that it is (perhaps the semantics of) stativization in stative passives that produces the contrast. It appears that certain types of modifiers are likewise incompatible embedded under the ‘stativization’ associated with English perfect participles; cf. *Beatrice has been to the store (*yesterday at 3p.m.)*. However, while certain types of modification might be in principle incompatible embedded in stativized structure in certain English participles, there does not appear to be a clear reason why this should be the case for the full range of XP modifiers in (25a-b). See Embick (2023), and Alexiadou et al. (2015); Ramchand (2018) for discussion.

Based on contrasts like those in (25a-b) and some additional observations, Embick (2023) argues that a small syntactic analysis of stative participles is to be preferred over both lexical and phrasal syntactic alternatives. What it means to be *small* in this sense is that the structure of the participle is built by what can be called (*direct*) *head merge*: an operation that attaches one head to another, with the output being a complex head. In the analysis in Embick (2023), *y* may be directly merged with *x* to create a complex head [*x y*]. This complex head, in turn, may have a head *z* merged with it to create the complex head [*x[xx y] z*]. However, if a phrase *z*P is merged, the output is phrasal, i.e. [_{XP}[*x y*] ZP].¹⁰

A stative passive participle constructed in this way is shown in (26); we simplify the participle structure here and in the remainder of our discussion by omitting Voice, as its status is not directly relevant to our analysis.¹¹

(26) *small stative passive*



The head α takes a verbal structure and creates a structure with a non-verbal distribution; that is, it

¹⁰For this kind of operation, see Embick (2004) for an application in stative passives, as well as Paparounas (2023). Wood (2023), Benz (2023), and Lee (2024) make use of small structures for certain types of nominalizations. For direct head merge itself, see Embick and Noyer (2001) in an analysis of English *do*-support. See also the discussion in Epstein et al. (2016); Moro and Roberts (2023); Oda (2024).

¹¹On this point, the status of Voice in stative passives is controversial; see, e.g. Anagnostopoulou (2003); Meltzer-Asscher (2011); Bruening (2014); Alexiadou et al. (2015); Bešlin (2023); and Paparounas (2023) for different perspectives. For the status of Voice in English stative passives, compounding of a type that will be examined in detail §5.1 may provide pertinent evidence. The relevant examples are as follows, with what appears to be an ‘incorporated’ Agent:

- (i) The mule-kicked football/ Susan-examined students/ robot-ridden dinosaurs ...
- (ii) The clown-amused children/student-grown azaleas ...

Crucially, the examples in (ii) show that an agentively interpreted argument is possible in participial Agent compounds even with Roots that are argued to not be associated with (Encyclopedic) agentivity, such as $\sqrt{\text{AMUSE}}$ and $\sqrt{\text{GROW}}$. This suggests that Voice is indeed present. There is likely much more to be said, however. Since our main claims do not depend on the presence or absence of Voice, we will not examine this matter further.

creates a participle that can occur in contexts that are typical of ‘adjectives.’ We use α for this head and not another label– e.g. a category-definer like *a*, or *Asp* for ‘Aspect’– to avoid extraneous associations that these labels inherit from other approaches. We also avoid calling α something like ‘stat(ivizer)’, because not all structures with this head are interpreted statively. We take up these points concerning the label in greater detail in §6.

To preview, one consequence of a small analysis is that in (26) the argument that appears with the stative passive cannot be introduced in a vP , where (under standard assumptions) a DP argument may appear as the complement of the verb. Instead, the argument that is modified by the stative passive must be introduced higher, outside the categorizer. We examine this point in detail in §4.

3.2 Modification from outside of α

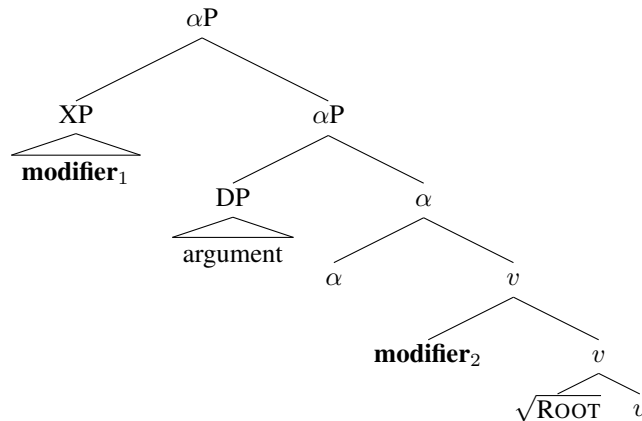
With the small structure (26) at hand, we are now in a position to explain the modification patterns identified at the beginning of this section. We first introduce some additional facts concerning manner adverbs, before synthesizing the general picture that emerges on the small analysis.

Manner adverbs have been widely discussed in the literature on stative passives. The facts to be explained are complex. In summary form, there are circumstances in which manner adverbs distinguish stative passives from simple adjectives, as in (27a). However, manner adverbs are not possible across the board with stative participles (27b). Finally, any unacceptability with stative participles as in (27b) must be explained relative to the fact that the same eventive adverbs are perfectly acceptable with the corresponding eventive passive (27c).

- (27) a. This box is carefully opened/*open. *stative passive/pure state*
 b. #This package is secretly/recently/diligently opened. *stative passive*
 c. This package was opened secretly/recently/diligently by the students yesterday. *eventive passive*

As a first step, we note that a small analysis of stative passives requires that phrasal modifiers be introduced above the stativizing head, i.e., they are introduced as **modifier**₁ in (28):¹²

- (28) possible modifiers (small account) *(to be revised)*



For completeness, we have included a **modifier**₂ position in the structure (28) to indicate that elements head attached below α are in principle possible in stative passives. This includes various types of prefix that may appear in the complement of α (e.g. *be-*, cf. *unbemoaned*, or *re-* in *unrepainted*) (though for

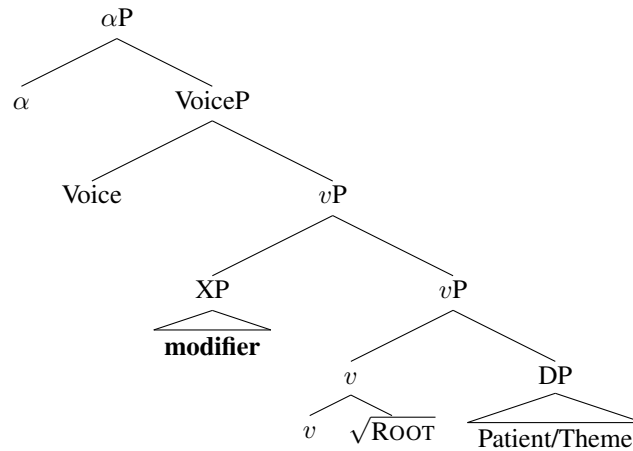
¹²The way in which the DP argument is introduced in (28) is provisional and will be revised in §4.

some of these, the term ‘modifier’ might be inappropriate). As affixal modification is not our primary focus, we put it to the side for the remainder of this work.

The small analysis in (28) establishes a straightforward structural explanation for (27b) versus (27c) by forcing an ‘external’ distribution of phrasal modifiers in the **modifier**₁ position. In the stative passive structure in (28), the XP modifiers cannot attach to a *v*P, because there is no phrasal projection of *v* under α ; the XP modifiers may only attach at α P. The manner modifiers in (27b) therefore do not have direct access to the event associated with the verbalizing structure, and the examples are consequently deviant (but see discussion later in this subsection for care on this point).

In the eventive passive, on the other hand, there is a full phrasal *v*P. XP modifiers in eventive passives may thus attach directly to it, as shown in (29):

(29) Eventive passive



The same explanation extends to the full range of modifier contrasts summarized in (25). In each case, XP modifiers are fine with eventive passives, where they attach to *v*P projections, but deviant in stative passives, where phrasal *v*P structure is absent.

The remaining complication to be explained is the fact that phrasal modifiers are not ruled out across the board in stative passives, as illustrated in (27a) (and the # in (27b)). The seemingly ‘mixed’ behavior of modifiers in stative passives has been investigated in Rapp (1997); Meltzer-Asscher (2011); Alexiadou et al. (2015); Gehrke (2015); McIntyre (2015, and references there). The findings of these studies are subsumed under the *State Relevance Hypothesis* in McIntyre (2015), whose formulation is as follows:

- (30) **State Relevance Hypothesis:** Event-related satellites are unacceptable in (German, English, Hebrew) adjectival passives unless they contribute to the description of the state expressed by the participle or of the theme during the interval *i* during which this state holds. They are most acceptable if they provide information which can be inferred solely by inspection of the theme during interval *i*.

In the remainder of this section we show that the statement in (30) covers a number of important observations that follow from how modification of stative passives must work on the small analysis. In particular, the requirement for a State Relevant interpretation of the manner modifier is precisely what is expected if XP adverbs attach outside of α P, as in (28), where they are interpreted as modifying a state. Note for now that the focus is on predicative syntax and @ interpretations; what happens with the E-interpretation is addressed in §3.3.

We begin by observing that State Relevance correctly describes the contrast between *carefully opened* and #*secretly opened* in (27a) and (27b). Whereas care in opening can straightforwardly be manifested

in an opened state of the package (e.g., if the package is not destroyed, or if the wrapping is folded and untorn), it is not at all clear out of context what it would mean for the secrecy of an opening event to manifest itself in a state in the same way. Extending this point, consider now a scenario in which it is known that there are two types of package-opening that can occur: one overt type of opening, and one secret type, and that these are known to produce distinct effects. Suppose, for instance, that typical overt openings involve ripping boxes open with the hands, whereas secret openings involve cutting tape with a knife, and careful unfolding of wrapping paper. In this context, it is now possible to coerce modification with the manner modifier *secretly* (in contrast to (27b)). In examining an open package with perfect paper next to it, (31) is now felicitous:

(31) Unlike those messy ones over there, this package is secretly opened.

Finally, State Relevance correctly extends to XP modifiers generally – not just manner modifiers. Connecting with the contrasts in (25), consider now the examples in (32), which are adapted from McIntyre (2013). Instrumental *with* in examples like (32a) seems odd without context; but knowledge of a particular effect produced by a brush makes the example felicitous. The same effect is found with an Agentive *by*-phrase in the example in (32b), where some property of the Agent is understood to be observable in the result state described by the participle:

(32) a. Bobi's hair is combed [with Susan's brush]. (# Out-of-the-blue)

But, SCENARIO: It is known that Susan's brush produces an amazing zigzag pattern.

b. These trees are scratched [by a badger]. (# Out-of-the-blue)

But, SCENARIO: We distinguish trees scratched by badgers versus trees scratched by bears based on the distance between the claw marks.

To summarize, State Relevance effects of exactly the kind just outlined are expected if all XP adverbs attach outside of α P in stative passives, as shown in (28). State Relevance effects are not expected if such phrases could attach directly to a v P. Thus, the complex observations collected under State Relevance provide further evidence in favor of the small analysis adopted here.

3.3 Modification and the attributive/predicative contrast

We conclude this section by showing that the distribution and interpretation of phrasal modifiers take on new significance in light of the attributive/predicative asymmetry identified in Section 2.

To start with, we observe that the facts at issue in §3.2 confirm that predicative stative participles lack an E-interpretation. If an E-interpretation were available for the predicative participle in (27b), repeated as (33), then the predicative stative passive should freely license manner readings of phrasal adverbs, just like the eventive passive. This is clearly not the case:

(33) #This package is secretly/recently/diligently opened. *predicative*

On the other hand, in §2 we observed that an E-interpretation is available when the stative participle is in attributive position. According to this generalization, there should be no deviance of event-related manner adverbs with attributive participles. This is correct:¹³

¹³Our focus is on manner adverbs because many other modifiers that could in principle be used are precluded from attributive environments due to independent properties of English DP-internal syntax. In particular, modifiers cannot have post-head material: *the flattened *with a wrench box* (cp. **the sick with the flu child*, etc.).

- (34) The secretly/recently/diligently opened package... *attributive*

Some further examples are given in (35) and (36); as expected, the felicity of the predicative examples can be manipulated by adding context:

- (35) Attributive
- a. The slowly/quietly/sloppily typed letter...
 - b. The surreptitiously flattened boxes...
 - c. The violently kicked ball...
- (36) Predicative
- a. #This letter is slowly/quietly/sloppily typed.
 - b. #These boxes are surreptitiously flattened.
 - c. #The ball is violently kicked.

Attributive participles can also have an @-interpretation. The prediction is thus that State Relevance effects should also be found for attributive participles as well, as long as they are interpreted statively. This appears to be correct. Recall the scenario employed in which it is known that secret openings have detectable consequences in the end state. It is then possible to have (37):

- (37) SCENARIO: There are ten open boxes on the table: 4 extremely tidy, and 6 with various marks etc.

The secretly opened boxes are the ones with the folded paper. @

Additional types of modification provide further confirmation for what we have proposed concerning the @- versus E- distinction and its distributional sensitivity. For example, proportional (scalar) adverbs such as *half* or *completely* naturally apply to states. On the other hand, frequency adverbs such as *twice* or *repeatedly* apply to events. Both state- and event-related adverbs are perfectly possible in the attributive position (38). With predicative participles, in contrast, state-sensitive modifiers are felicitous, while event-sensitive modifiers are not (39).

- (38) Attributive
- a. State type: The half-/completely painted desk ... @
 - b. Event type: The twice/repeatedly painted desk ... E
- (39) Predicative
- a. State type: This desk is half-/completely painted. @
 - b. Event type: #This desk is twice/repeatedly painted. *E

It is possible to coerce readings for the predicative participles with respect to State Relevance as in (39b), and so again these examples are marked #. This is most obviously the case if having been painted twice or repeatedly leaves a detectable mark on the painted desk, – accumulated paint, for example. It should be clear that on the felicitous reading of (39b) the adverb is modifying an @-state.

Another diagnostic that distinguishes the two interpretations is modification with *almost* (see e.g. Rapp and Von Stechow, 1999). As Paparounas (2023) observes, *almost* has a scalar interpretation with @-states; on the other hand, with events like the E-interpretation, *almost* is counterfactual, with the assertion being that an event did not take place. As expected, *almost* @-/E- ambiguity is found with attributive participles (40). In contrast, no such ambiguity is found with predicative participles (41); only the state-related @-interpretation is possible.

(40) Attributive: The almost flattened box...

@: almost attained end state;

E: almost got flattened (counterfactual)

(41) Predicative: This box is almost flattened.

@: almost attained end state;

*E: almost got flattened (counterfactual)

To summarize, event and state modifiers, *almost* modification, and the behavior of State Relevance in attributive syntax together confirm our generalization about stative participle interpretation and its distributional sensitivity to attributive versus predicative syntax.¹⁴

3.4 Interim conclusions

This section has established three points. First, XP modifiers of stative passives attach above the α head. There is no possibility of attaching an XP modifier to a *vP* in a stative passive, in contrast to active or eventive passive verb phrases. Second, XP modification of stative passive participles must meet the conditions of *state relevance* if the participle is interpreted as an @ state. Third, we have proposed that attributive participles have an E-interpretation, and that in this case State Relevance effects do not apply. The small analysis provides a structural account for all the effects involving phrasal modification in a straightforward way; alternative phrasal analyses of stative passives do not.¹⁵

4 Category and argument introduction

We now turn to how stative passives are interpreted, concentrating on why stative passive participles have different interpretations and why these are sensitive to attributive versus predicative syntax.

The analysis to be developed has three components. The first requires an examination of what it means for participles to be *deverbal* in the sense associated with α . The second concerns how arguments are introduced in an ‘abstract’ theory of syntactic category. The third involves contextual interpretation, of a kind that is possible in a theory of abstract category; what has been called *allosemy*. In our implementation, allosemy allows for one and the same participle-defining head to be interpreted differentially depending on the local syntactic context: sometimes α is interpreted statively, sometimes it is not.

After identifying the full range of what is to be explained in §4.1, we will introduce the key ingredients of the analysis in §4.2, and implement it in §4.3, where the core conclusion is that English passive participles can be interpreted eventively only when they do not introduce the argument to which they apply. §4.4 provides a summary.

4.1 The full range of what is to be explained

The full set of generalizations to be accounted for becomes clear when further aspects of the eventive passive participle are considered. This participle provides crucial evidence on the question of how α is

¹⁴We have not addressed modification with *still* in our discussion, a modifier that has often been appealed to as identifying certain types of (Target) state interpretations in prior work (Kratzer, 2001; Anagnostopoulou, 2003; Alexiadou et al., 2015, and references there) (following Nedjalkov and Jaxontov, 1988). On the limitations of *still* as a diagnostic, see Gehrke (2012); Irmer and Mueller-Reichau (2018); Baglini and Kennedy (2019); Paparounas (2023).

¹⁵In the interests of space we have not argued against lexical accounts, which make predictions about modification that are essentially the same as the small approach. For distinguishing the small and lexical account see Embick (2023), and §4 for some comments.

interpreted semantically. To see this, we start with a look what the eventive and stative participles share, and how they differ.

Starting with shared properties, a first is that the eventive passive participle is invariably identical in form to stative passives, in ways that include unpredictable (=Root-determined) allomorphy. This observation figures prominently in previous analyses of English participles (e.g. Levin and Rappaport (1986), as well as Embick (2003, 2004)).

A second shared property is that eventive and stative participles have essentially the same syntactic distribution; see Freidin (1975); Bresnan (1982); Embick (1997); Emonds (2006, 2007, 2017); Lundquist (2013); Bešlin (2023). To take one example, the English passive has the syntax of the typical copula construction in predicative syntax, as discussed extensively in Embick (1997). For this reason, English passives license expletive *there*, as in the eventive passive (42b):¹⁶

(42) predicative syntax for the eventive passive

- a. Six people were arrested by the police at two p.m. yesterday outside of the store.
- b. There were six people arrested by the police at two p.m. yesterday outside of the store.

A final key property concerns argument interpretation. The argument of the stative passive with an E-interpretation is an event participant (Patient/Theme), exactly the same as in the eventive passive (and, for that matter, the corresponding active):

- | | | |
|------|---|-------------------------|
| (43) | a. The violently kicked box ... | <i>stative passive</i> |
| | b. The box was kicked violently by the barista. | <i>eventive passive</i> |
| | c. The baristas kicked the box violently. | <i>active</i> |

One crucial way in which eventive and stative participles differ is that eventive passive participles in predicative syntax are always interpreted eventively. This is not the case for stative passive participles, which lack the E-interpretation when used predicatively.

Taken together, the generalizations to be explained are summarized in (E1-3):

- (E1) The eventive passive participle and stative passive participle have an identical form, and the same distribution.
- (E2) The stative passive can have an @ interpretation in both attributive and predicative positions.
- (E3) The eventive passive is interpreted eventively in predicative syntax. The stative passive has an eventive interpretation only in attributive syntax.

4.2 Components of the analysis

We start from the idea that the syntax of the two participles is defined by the syntactic head α (as introduced in §3); in particular:

¹⁶ The eventive passive participle may also occur DP-internally. This is most straightforward in post-nominal ‘reduced relative’ syntax, as in (16c); the adjectives in (16a,b) illustrate the connection between the attributive and post-nominal position.

- (i) a. The [sick] students stayed home.
- b. The students [sick with the flu] stayed home.
- c. The boxes [recently flattened/kicked by the baristas] ...

The distribution of eventive passives in attributive syntax raises important questions that are addressed in detail in §5.

(44) α : head that determines a non-verbal distribution.

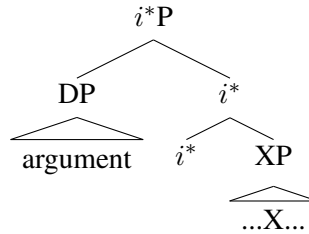
A *verbal distribution* refers to the types of functional syntax that is found in typical verbal clauses, where v might be dominated by Tense (and perhaps other heads). The presence of α defines a different type of distribution. This means that the syntactic object headed by α has the properties of α , and not of the $v(P)$ it dominates. More specifically, we assume that categories like α define an *extended projection* when additional functional material of the correct type is merged above them; see Grimshaw (1991, 2005) and Marantz (2022). Other heads may appear above α in its extended projection, in the same way that various heads may appear above v in verbal clauses or above n in the extended projection of a nominal. When an α phrase appears in predicative syntax, it is worth noting that it occurs with a copula that combines with Tense: *The boxes **are** flattened.*

In our analysis, α appears in both stative and eventive passives. The fact that they share some properties and differ for others arises from the assumption that syntactic categories are abstract: a single category does not define a unique way of combining with arguments or a single semantic interpretation. We examine each of these points in turn.

4.2.1 Argument introduction

We assume a generalized view of external arguments according to which they are introduced by the head i^* (Wood and Marantz (2017); on the Marantz (2022) view all arguments are introduced by heads of this type), abstractly schematized in (45):

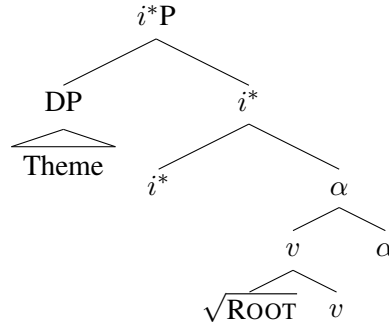
(45) i^* : head that introduces (external) arguments



The argument introduced by i^* is interpreted in relation to the complement of that head. So, for example, when i^* occurs with a vP complement, it is essentially what is called a *Voice* head (cf. Kratzer (1996) and subsequent work). In this case, the argument introduced in its specifier is (typically) interpreted as an Agent. To avoid confusion, we will use the label *Voice* when referring to this configuration of i^* .

In the stative passive, the argument introduced by i^* is a Theme (i.e. a holder of a state). The Theme argument must be introduced by i^* because the small structure precludes it from being introduced within the complement of v : phrasal argument introduction would require a phrasal vP that the stative passive cannot contain a vP , as shown in §3. Instead, the stative passive argument must be introduced externally to α , by i^* , as in (46):

(46) Argument introduced in stative passive



Stative passive arguments have been argued to require external introduction in many prior studies; for different perspectives, see Levin and Rappaport (1986); Horvath and Siloni (2008); Meltzer-Asscher (2011, 2012); McIntyre (2013); Alexiadou et al. (2015); Biggs (2021); Paparounas (2023). Different kinds of evidence support the external introduction. We outline two here, leaving a more detailed treatment for another occasion; see also Paparounas (2024b).

One type of evidence comes from the Root-specific effects that we have seen at various points above; for example, the fact that *The boxes are flattened* is felicitous, while *#The boxes are examined* is not. This behavior makes sense given (46), in which all arguments introduced by i^* in stative passives are Themes. While *flatten* is easily understood as occurring with such arguments, *examine* is not; hence the contrast. On the other hand, it is not clear that this behavior is expected in accounts that introduce the stative passive argument in a typical vP .¹⁷

Further evidence supporting external introduction comes from the scope of negation. First, by way of comparison, we note familiar scope ambiguities between the derived subject and negation in eventive passives (47a), and that stative passives with sentential negation show the same behavior (47b).

- (47) a. Every door wasn't opened. $\forall > \neg, \neg > \forall$
b. Every door isn't opened. $\forall > \neg, \neg > \forall$

The scope of sentential negation in (47b) contrasts with affixal negation in stative passives. The stative passive with *un-* in (48) lacks the ambiguity found with sentential negation in (47); *un-* negation only has the narrow scope interpretation:

- (48) Every door is unopened. $\forall > \neg$

We take it that English sentential negation merges at a phrasal level, while affixal *un-* attaches directly to a head: i.e., to α in the small stative passive; see also §5.2. The ambiguity in (47b) follows if the QP originates in the specifier of i^* in the stative passive, i.e. below negation. The absence of a reading with the QP in the scope of negation in the stative passive in (48) follows from the small analysis: since the DP is introduced in the specifier of i^* , external to α , the DP merges above negation and outside of the scope of *un-*. Again, the contrast between (47a-b) and (48) is not explained by phrasal accounts (or, more generally, by accounts that introduce the stative passive argument in the same way as a typical vP argument), in which there should be no difference between the eventive and stative passive.

¹⁷This amounts to a kind of Theme analysis of the stative passive, which has a long history; see e.g. Anderson (1977). On the general theme of argument introduction, stative and eventive passives also differ with respect to whether certain types of arguments are possible; notably with respect to double objects (Wasow, 1977; Levin and Rappaport, 1986). For how this might work on a small analysis, which excludes arguments that require introduction via phrasal structure, see the preliminary discussion in Embick (2023).

4.2.2 Contextual interpretation

Abstract morphemes are interpreted at both the form and meaning interfaces in ways that may be influenced by the syntactic context in which they occur: there might be contextual allomorphy on the form side, or allosemy on the meaning side.

At the PF interface the α head receives the same form in the two participle types (Embick, 2003, 2004); that is:

(49) spell out of α

$$\begin{aligned}\alpha &\leftrightarrow \text{-en/}_\{\sqrt{\text{BREAK}}, \sqrt{\text{SPEAK}} \dots\} \\ \alpha &\leftrightarrow \text{-}\emptyset/_\{\sqrt{\text{HIT}}, \sqrt{\text{SING}} \dots\} \\ \alpha &\leftrightarrow \text{-t/}_\{\sqrt{\text{BEND}}, \sqrt{\text{BUY}} \dots\} \\ \alpha &\leftrightarrow \text{-ed}\end{aligned}$$

The fact that stative and eventive passive participles are always identical formally is accounted for by the presence of α in both structures, in conjunction with the Vocabulary fragment in (49).¹⁸

With respect to meaning, the particular type of contextual effect that we propose is stated in (50), which specifies both (i) how to interpret the argument introduced by an i^* attached to α , and (ii) that α in this scenario receives a stative interpretation:

- (50) a. Syntax: [i^* [α ...]]
 b. Semantics:
 i. Interpret α as State; and
 ii. Interpret a DP introduced by i^* as a Theme (i.e. a State Holder).

The interaction of i^* and α specified in (50) is a type of allosemy; see (cf. Marantz, 2013; Wood, 2023; Myler and Marantz, 2024) for different perspectives and applications. When α occurs as the complement of i^* as in (50a), α is interpreted as a state, as specified in (50a). When α appears on its own (without i^*), α has a syntactic consequence– it determines a nonverbal distribution– but it has no semantic interpretation. We will examine this use of allosemy with respect to other uses of this notion in §6.

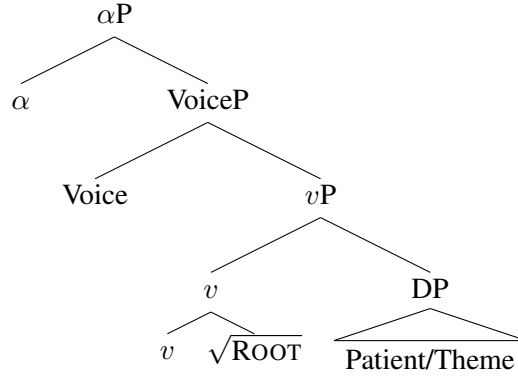
With these proposals at hand, we will now show how they account for the generalizations identified above.

4.3 Application

Our analysis of the eventive participle holds that it is built with α , like the stative passive, but with this head taking a phrasal complement:

(51) Eventive passive

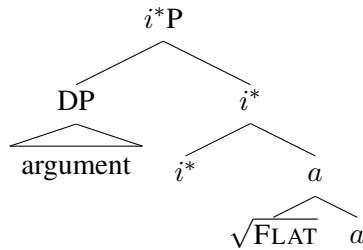
¹⁸There are ways in which the Vocabulary fragment in (49) would require modification in a more comprehensive analysis of the verbal system: the *-ed* realization of α is also the default realization of T[+past] in English; similar considerations apply for the *-t* and *-* $\emptyset$ allomorphs. It is conceivable that such an analysis would make reference not to α per se, but to some other feature common to both Tense and what is typically thought of as Aspect. We leave this matter for future research.



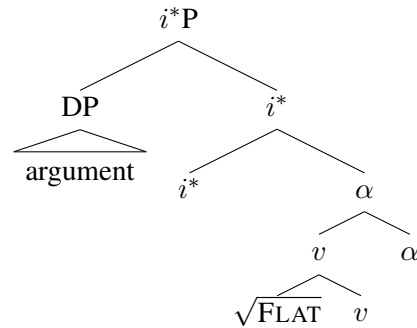
The phrasal complement is motivated by the fact that the eventive passive behaves like it contains a typical vP , unlike the stative passive; see §3. We assume that the eventive passive participle is built by merging α above VoiceP.¹⁹ In syntax, the α head defines a non-verbal distribution. It is for this reason that eventive passive participles have the (predicative and attributive) distribution of adjectives, and not the distribution of verbs. Affixation in (51) produces a complex head $[[[\sqrt{\text{ROOT}}] \text{Voice}] \alpha]$ in which α is realized by the participial exponents *-ed*, *-en*, etc. via (49). Taken together, the previous two observations account for (E1): the identical form and distribution of stative and eventive passive participles. Finally, with respect to meaning, α is not embedded under i^* in (51); thus, by (50) the α head does not receive a semantic interpretation. Instead, the interpretation of the eventive passive participle is that of a typical VoiceP; i.e., it is eventive.

The small stative passive and ‘normal’ adjectives (*red*, *tall*, etc.) are also syntactic objects that can appear in predicative position. Illustrating each of these with the Root $\sqrt{\text{FLAT}}$, the structures are as follows:

(52) adjective *flat*



(53) stative passive *flattened*



In both structures, the argument is introduced externally, by the argument-introducing head i^* . We address further possible connections between structures with a and α in §6.

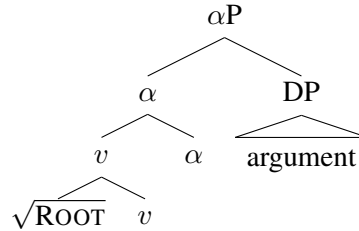
In the stative passive in (53), α defines a non-verbal distribution and its participial form in the same way as in the eventive passive. However, in contrast to α in the eventive passive, α has an interpretation in (53), due to the local i^* morpheme. In the formulation of allosemy that we have proposed, the interpretation of the participle is necessarily stative when α is the complement of i^* .

One point that follows from the small analysis and external argument introduction is that the i^* morpheme configuration in (53) is required when the stative passive occurs in predicative syntax. Without

¹⁹We ignore the status of the Agent argument because it is not relevant to our main claims.

this additional syntactic head, no argument can appear in predicative syntax since (to repeat) there is no phrasal vP in the stative passive in which the argument of the participle could be introduced. External argument introduction also means that an argument cannot be a direct complement to α , as in (54):

(54) *not possible*: α is not an argument introducer

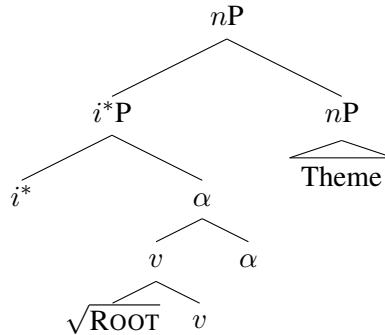


The intuition prohibiting (54) is that α (in contrast to e.g. v) cannot be *transitive*, i.e., introduce an argument ‘directly’ (=without an additional argument-introducing head).²⁰

If (53) is possible and (54) is not, there is an explanation for the component of (E3) that specifies that stative passive participles are not interpreted eventively in predicative syntax. In predicative position, the stative passive must have its argument introduced by i^* ; given the obligatory presence of i^* when the participle occurs predicatively, the interpretation is necessarily stative.

The structure in (53) may also appear as an attributive modifier; this is illustrated in (55):

(55) Attributive, α under i^*



In exactly the same way as in (53), α is interpreted as a state in (55) due to the presence of i^* . The fact that the same participial structure with i^* and α can occur in both attributive and the predicative syntax accounts for (E2): stative passive participles may be interpreted as @-states in both syntactic positions.

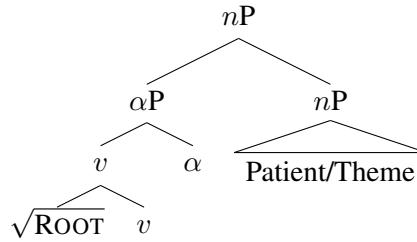
What remains to be explained is the second part of (E3), *What is it about attributive stative passive participles that allows them to be interpreted eventively?* Our answer to this question is shown in (56), which is the result of directly adjoining a stative passive αP to an nP for attributive modification. Crucially, there is no i^* in this structure.²¹

²⁰Many analyses of the category *adjective* share this assumption; see Bešlin (2024) for discussion. For one that employs (generalized) i^* , see Marantz (2022). While a ban against direct introduction of DP arguments by α is clearly needed, it is important to note that participles and adjectives do appear with different types of post-head phrases that are often assumed to be complements: e.g. *Magnus remained unpersuaded that this was the best option*, or *Romaine is certain to have a decent outing*. See Benz (prep) for analysis and discussion of differences in the behavior of DP and clausal complements across categories.

²¹We take it that the stative passive structure $[[\sqrt{\text{ROOT}} v] \alpha]$ is ambiguous in constituent structure: $\alpha(P)$, along the lines of Chomsky (1995). Adjunction of the head (α) to a phrase (NP) as in (56) makes α phrasal (under the assumption that heads may not be adjoined to phrases).

Note in addition that the object in (56) may appear in predicative syntax, but the interpretation will be that of a modified noun, presumably what is found in copular sentences like *That is a [flattened box]* (cp. *Mary is a [happy person]*).

(56) Attributive, attaching α P to a n P



On the proposal we have advanced, α is not stative in (56); the interpretation is thus determined by the complement of α . In (56), this means that the α -headed participle is interpreted as an event that can have a Patient event participant, because the [v $\sqrt{\text{ROOT}}$] that α merges can be interpreted in this way. The key aspect of this proposal is that in (56) the α participle does not introduce an argument itself; rather, the participle provides an interpretation for an **independently introduced** argument. Thus, in (56) α P finds such a nominal in a configuration in which the n P that it is attached to is interpreted as a Patient. That the n P argument with which the participle merges is independently licensed is clear from the fact that the participle (like an adjective) is completely optional: *the (flat/flattened) boxes...*; cp. **The boxes are*.

To summarize, attributive participles may attach to n Ps either with i^* , so that they are interpreted statively (55); or without i^* , with an eventive interpretation (56). In predicative position i^* is required to introduce the participle's argument (as in (53)); as a result, such participles are always interpreted as stative. This completes our explanation for (E2-E3).²²

²²The analysis in the main text could potentially be linked to some larger questions in the theory of modification. Having the participle attach to n P in the ways shown in (55) and (56) produces two compositional puzzles that connect with more general questions in the semantics of adnominal modification. The first concerns the semantic types required for attributive (vs. predicative) modification, and is an issue shared by essentially all treatments of attributive adjectives and stative participles. The second puzzle is specific to our approach and concerns the E-interpretation in examples like *The (recently) kicked boxes...*

On the first point, it has long been recognized that there is an apparent type mismatch for attributive modifiers and the NPs that they attach to. See, for example, Morzycki (2016) for an overview. On the second point, our approach requires that attributive modifiers might compose with the noun in more than one way. Specifically to be addressed for the E-interpretation is that the participle is interpreted as its v complement, i.e. as a transitive verb that takes a Patient argument. However, while a typical verb combines with a DP, the participle here is combined with an n P, which is of a different semantic type.

Another interesting connection between our discussion of stative passives and attributive modification more generally is that variability in adjective-noun modification semantics has long been known to be sensitive to the syntax of the adjective. For example, many non-intersective adjectives in English are restricted to a (prenominal) attributive distribution (*alleged, probable, likely, former, etc.*), as in (i). In addition, for adjectives that can be ambiguous between an intersective and non-intersective semantics (*old, beautiful, etc.*), the non-intersective interpretation is similarly restricted to the immediately prenominal attributive position; see (ii).

- (i) a. *The chef is former/ future/ erstwhile/ alleged/ probable/ occasional.
b. The former/ future/ erstwhile/ alleged/ probable/ occasional chef...
- (ii) a. a beautiful creative dancer
dancing is creative (#beautiful dancing)
b. a creative beautiful dancer
dancing is beautiful (#creative dancing)

Many further distributionally-sensitive interpretations are discussed in the attributive adjective semantic literature, including differences between 'direct' and 'indirect' modification, restrictive and nonrestrictive modification, and stage and individual level interpretations, as has been discussed since at least Bolinger (1967). Again, while the relevant phenomena are not uniform in terms of the semantics, what unifies such effects is their restriction to attributive syntax in English. One goal for future work is to investigate how the behavior of participles might connect with these other facts implicating DP syntax and semantics.

4.4 Summary

The analysis presented in this section is centered on syntactic structures that share the head α . These are the eventive passive, in which α takes a phrasal VoiceP complement, and the stative passive, where the complement of α is small. We have further distinguished between whether α is the complement of an i^* head. Finally, we argued that the small stative passive cannot appear in predicative position without i^* . Taken together, the analysis produces (57):

(57) Summary table

structure	name	predicative	attributive
$[\alpha [\text{VoiceP} \dots]$	eventive passive	E	E
$[\alpha [v \dots]$	stative passive	*	E
$[i^* [\alpha [v \dots]$	stative passive	@	@

The analysis summarized in (57) accounts for (E1-E3) in the following ways:

- (E1) *The eventive passive participle and stative passive participle have an identical form, and the same distribution.* $\Leftarrow$ Both participles have their syntactic distribution and form defined by α .
- (E2) *The stative passive can have an @ interpretation in both attributive and predicative positions.* $\Leftarrow$ The stative passive may have an i^* that introduces a state holder argument in both of these environments, resulting in the interpretation of α as a state.
- (E3) *The eventive passive is interpreted eventively in predicative syntax. The stative passive has an E reading only in attributive syntax.* $\Leftarrow$ The eventive passive does not require i^* above α to introduce an argument; it is thus always eventive. A stative passive without i^* may be attached to an independently-licensed DP as a modifier; in the absence of i^* it is interpreted in the same way as its $[\sqrt{\text{ROOT}} v]$ complement, i.e. eventively.

In concluding, we note that our analysis relates the stative and eventive passive through the head α , but does not derive the stative passive from the eventive passive. A derivational relationship has been put forth previously in both syntactic (e.g. Bruening, 2014) and lexical (e.g. Levin and Rappaport, 1986) approaches. With respect to the former, the arguments developed in §3 provide clear evidence against deriving statives from eventives in syntax. If the stative passive were derived syntactically by attaching a head above the eventive passive, then it is predicted that these two participle types should behave identically for modification (and argument introduction), contrary to fact. There is an interesting comparative question that arises on this point. Analyses of Modern Greek by Paparounas (2023, 2024a), Ardalani Kurdish by Hamo and Meihami (2024), and Brazilian Portuguese by Lopes and Biggs (2024) have argued for smaller structures, as we have for English. Bešlin (2023, 2024), however, has recently argued that the stative passive in Bosnian-Croatian-Serbian (BCS) has internal vP structure. One objective for future research will be to characterize and explain cross-linguistic variation (see also §6).

Lexicalist analyses like Levin and Rappaport (1986) (following Bresnan, 1982, among others) also derive the adjectival (=stative) passive participle from the verbal (=eventive) one. Lexical analyses have an interesting property in connection with our approach: lexical analyses predict no vP -attached phrasal modifiers in stative passives—since they posit no vP to attach to—just as small analyses do. This raises the question of whether the predictions of the lexical and small analyses can be distinguished from one another. Embick (2023) argues that they can based on the interaction of *un*-prefixation with stative passives and resultatives. In brief, it appears that a lexical account has no principled explanation for why lexical rules cannot produce *un*-prefixed participles that take resultative secondary predicates: e.g. **The*

metal is unhammered flatter than a pancake, where the version without negation is grammatical. A small account correctly rules out this type of example, in ways that involve details beyond what we can discuss here. What other types of phenomena or interactions provide evidence either way remains to be seen.

5 E-interpretations: Stative and eventive participles

We have argued that there are two structurally distinct passive participles in English that have an eventive interpretation. One is the stative passive without i^* , and the other is the eventive passive participle. Given that our analysis posits two participles that are identical in form and interpretation, the question to be addressed is whether it is in fact necessary to posit a small participle with an event interpretation. A plausible alternative hypothesis is that the E-interpretation of the participle simply derives from an attributive eventive passive participle in examples like (58):

- (58) a. The recently kicked boxes...
 b. Some carefully examined books...
 c. A flattened tomato...

Certain additional assumptions are required for an eventive passive analysis of prenominal attributive participles like those in (58). For example, with respect to argument interpretation, it would have to be assumed that there is an operator present in the participle phrase in (59a), along the lines of the postnominal participle in the ‘reduced relative’, illustrated in (59b):²³

- (59) a. The $[OP_i$ recently arrested $t_i]$ truckers ...
 b. The truckers $[OP_i$ recently arrested t_i (by the police)]

Nothing that we are aware of excludes the possibility that the participles in simple examples such as (58) are prenominal eventive passives. At the same time, nothing excludes the possibility that the participles in (58) are stative passives. Thus, it might be that either of two different syntactic objects can appear in the same syntactic position, with the same form and same semantic interpretation. Unless there are reasons for ruling the eventive passive participle out of the prenominal attributive position, there is a real question as to whether E-interpretation are truly associated with stative participles.²⁴

In its strongest form, the hypothesis to be tested in this section is (60):

- (60) HYPOTHESIS TO BE EXAMINED: E-readings in attributive position are *always* produced by Eventive Passive participles.

Our findings on this question are negative: (60) does not appear to hold across the board, and therefore our conclusion that there are stative participles with an E-reading appears to be correct. However, it is possible to verify this point only in highly controlled cases. What is required is a demonstration that particular participles (i) must be stative passive participles (and not eventive passives), and (ii) that these have an eventive interpretation. We center our discussion on two cases that have exactly these properties. The first involves compounding, and is addressed in §5.1. The second, involving negation with *un-*, is the topic of §5.2.

²³For various perspectives on reduced relatives as adnominal modifiers, see Ross (1967); Kayne (1994); Belk (2017); for discussion of participial attributive modification in reduced relatives, see Bhatt (1999); Iatridou et al. (2001); Marvin (2003); Stanton (2017).

²⁴See Meltzer-Asscher (2011, fn.1), as well as Sleeman (2011) for the idea that eventive passives have this distribution.

5.1 Evidence from compounding

English stative passives allow a kind of compounding that we refer to as *Agent compounds*; these were introduced briefly in Fn. 11 above. Further examples are given in (61):

- (61) a. The mule-kicked football flew off into the distance.
b. The Susan-examined students reported a positive experience.
c. The frost-touched window let in a charming light.

It is crucial that in the structures in (61), the incorporated nominals can have the same interpretation as the Agent argument in corresponding actives or *by*-phrase passives: compare *The mule kicked the ball*, *The ball was kicked by the mule*. We additionally facilitate an E-reading in these compounds by employing Roots that typically do not have good @-states.

The grammaticality of the attributive participles of Agent compounds in (61) is of interest because incorporation of an Agent is impossible with eventive passive participles. The participial compounds in (62), counterparts of (61) (using the same verbal Roots), are ungrammatical with the Agent interpretation of the incorporated noun:

- (62) a. *The football was mule-kicked yesterday.
b. *The students were Susan-examined at the meeting.
c. *The windows were frost-touched during yesterday's storm.

The relevant judgment for (62) is that in which the non-head nominal is interpreted in the same way as the *by*-phrase argument of the corresponding eventive passive, as in (63) (or the Agent of the active):

- (63) a. The football was kicked by a mule yesterday.
b. The students were examined by Susan at the meeting.
c. The windows were touched by frost during yesterday's storm.

We stress this point because examples like those in (62) have a grammatical interpretation that derives from the freedom with which English verbs allow compounding in verbs: *ninja-walk*, *pan-sear*, *Chomsky-adjoin*, and so on. In this kind of example, the noun is not interpreted as an argument of the verb (e.g. Roeper and Siegel, 1978). Instead, the noun specifies (to a first approximation) a manner or an instrument involved in the verbal action, i.e. in examples like *Rodrigo pan-seared the snapper*, Rodrigo is the Agent, and the searing takes place in a pan. The examples in (62) are acceptable to the extent that speakers accept the manner or instrument interpretation of the nominal. In (62a), for example, someone would kick the football the way a mule does (e.g. with a leg-moving-backward motion). That the manner/instrument interpretation is distinct from the Agent reading is evident from the fact that the eventive passive examples in (62) are compatible with the addition of a *by*-phrase specifying the Agent who performs the event in that manner: e.g. *The footballs were mule-kicked by the players*.

With the contrast between Agent compounding in eventive and stative passives in hand, we next establish a context in which to test the readings of these compounds.²⁵

²⁵Many other diagnostics involving pre-participle modifiers cannot be used with compounded forms. For example, Agent incorporation appears to be incompatible with further adverbial modification in English:

- (i) *The violently mule-kicked football flew off into the distance.

Why this should be the case is not entirely clear to us. Extant analyses of this kind of compounding, such as Roeper and Siegel (1978), propose that the participle contains a single position that can be the target of the lexical operation that produces a compound; but it is not clear what explains this restriction.

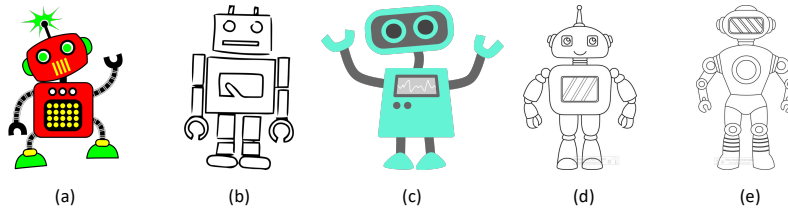
One context in which the @- and E-interpretations have distinct truth conditions exploits the observation that states may be *transitory*, in the sense that the state may cease to hold. In particular, certain @-states that obtain at time τ_0 may cease to obtain at a subsequent time τ_1 . As illustration, we use the Root $\sqrt{\text{PAINT}}$, and an initial context in which a set of robots may be painted at time τ_0 , as in (64):

(64) All of the new robots are painted.

The robot context can be updated such that subsequent action— weather effects, wear and tear, etc.— produces a transitory reading, in which the paint might come to be no longer present on the robots. For this update, we can say that it is true at the later time point τ_1 that the robots have undergone a painting event, but not that they are in a painted state.

The initial and updated robot contexts establish circumstances in which the @- and E-interpretations have different truth conditions. Consider now the five robots in (65):

(65) 5 robots



We employ the clauses in (66a-b) to probe the available readings of the participle in predicative position in (66a) and attributive position in (66b). Given only what is visible in (65), both questions are reasonably answered with (a) and (c). These are @-interpretations, since the state holds at the time at issue:

- (66) a. Which robots are painted? (a), (c) *predicative: @*
b. The painted robots are... (a), (c) *attributive: @*

Next, and updating the initial context to introduce a transitory scenario, imagine now that we have knowledge that robots (a), (c) and (d) have been painted in the recent past, but that the paint on (d) has completely peeled off. Under these circumstances, the attributive participle in (67b) can have two distinct answers, depending on whether the participle is interpreted as @ or E. Thus, in addition to the state reading of the attributive (67b.i), an additional option (d) is a possible response (67b.ii); this is the E-reading. As is typically the case, an adverb like *recently* facilitates the E-interpretation in (67) (but is not required):²⁶

- (67) a. Which robots are recently painted? (a), (c) *predicative: @*
b. The recently painted robots are...
i. (a), (c) *attributive: @*
ii. (a), (c), (d) *attributive: E*

Having identified both a syntactic object that is possible only with the stative participle (Agent compounding), and a context in which @ and E interpretations have different truth conditions (painted

²⁶Note in addition that coercion is required to interpret *recently* in predicative (67a), as is expected given State Relevance.

robots), we can test the hypothesis (60). To do this, we adapt the context to one in which it matters who did the painting. With respect to (65), if students paint robot (a) and teachers (c), it is truthful to utter (68):

(68) The student-painted robot is robot (a).

The participle in (68) could in principle have either the @- interpretation (assuming that we know something about the students' painting, e.g. that they prefer red) or an E-interpretation. To further isolate the E-interpretation, consider now a context in which teachers paint (c), while students paint (a) and (b). Subsequently, the paint on (b) peels off. It is then possible to state (69), where only the E-interpretation is true:

(69) The student-painted robots are (a) and (b).

The example in (69) is false on the @-state interpretation, which requires both robots to be in the painted state at Topic Time. The fact that (69) can be true in the relevant context thus shows that there is an E-reading for the Agent compounded participle, which— as we showed – must be a stative participle and cannot be an eventive passive.

5.2 Evidence from *un*-prefixation

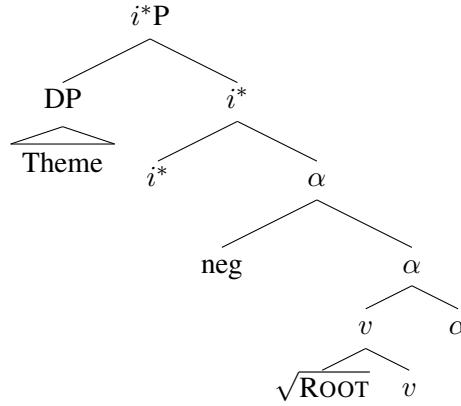
Further evidence that the stative passive participle can have an E-interpretation comes from *un*-prefixation.

The negative head realized as *un*- is relevant to participle formation in ways that have been noted since the early literature on the topic (cf. Siegel (1973)): *un*- has a highly restricted distribution with adjectives (70a) and participle-like forms as in (70b) (what are called 'Pure Statives' in Embick (2003, 2004)); in contrast, *un*- can always combine with the stative passive participle, as in (70c):

- (70) a. *unopen
 *unflat
 ⋮
 cp. unhappy, etc.
 b. *unlearnèd (cp. unlearned)
 *unblessèd (cp. unblessed)
 *unsunken (cp. unsunk)
 ⋮
 cp. unshaven
 c. unflattened
 unbeaten
 unkicked
 ⋮
 I like to blick apples. The unblicked apples.../These apples are unblicked.

As discussed in §4, the affixal Neg head realized as *un*- is directly head-attached to α (cf. Embick, 2023). This is illustrated for the stative passive in (71):

(71) *Negated participle, with i^**

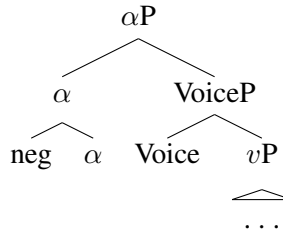


The freedom with which *un-* combines with the stative passive contrasts starkly with the eventive passive, which disallows *un-* completely.²⁷

- (72) *Mary was unarrested (by Tim)/unchased (by a wombat)/unphotographed (by a professional).

In our analysis, negation of the eventive passive is deviant for structural and semantic reasons. We assume that *un-* scopes over the structure that it c-commands. In the same way as in (71), in the eventive passive affixal *un-* can head-attach to α . However, as illustrated in (73), when *un-* attaches to α in the eventive passive, *un-* only scopes only over α , which (recall §4) has no semantic interpretation in this structural context:

- (73) *Negated eventive passive participle*



We take it that having Neg scope over nothing results in semantic deviance.

Negative *un-* robustly distinguishes stative from eventive participles. The next task is to probe for an E-reading for an *un-*-prefixed stative participle. What we find again confirms the attributive/predicative asymmetry: E-interpretations are possible with *un-*-prefixed participles, but only in attributive position.

We start with the simple observation that it is grammatical to attach *un-* to (attributive) participles like those in (74), which are constructed from verbs that lack good end states, and so are more naturally associated with E-interpretations rather than @-interpretations:

- (74) The unkicked boxes/untouched pencils/unexamined third option...

We further confirm that E-interpretations are possible with *almost* modification. If negated participles have an E-interpretation, it should be possible for *almost* to be interpreted counterfactually. This is indeed

²⁷Eventive passives are possible with verbs that allow ‘reversative’ *un-*, as in *Roger unbuttoned the sweater* ~ *The sweater was unbuttoned by Roger* (Siegel, 1973; Wasow, 1977; Bresnan, 1978); this interpretation is not the typical one for negation (though cf. Nissenbaum 2018 for an interesting discussion). For the stative participles in (70c) above (and throughout this section), note the impossibility of reversative inputs **unbeat* or **unkick*.

the case, as shown in (75):²⁸

(75) The almost unkicked boxes/ almost unwritten novel/ almost unexamined third option

The participles in (75) have a clear negated E-interpretation: e.g., that the boxes nearly went unkicked, but in the end finally underwent a kicking.

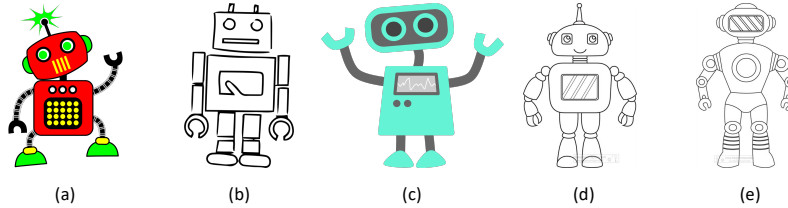
For completeness, the examples in (76) confirm that the interpretation of *almost* varies according to syntactic position in the ways expected with Roots with natural end states, here $\sqrt{\text{MELT}}$. To facilitate the degree reading, it helps to have a context in which we are seeking the ingots that are the most solid, i.e. which have been least affected by the melting process:²⁹

(76) modification with *almost*

- a. The ingots in Storeroom B are almost unmelted.
 - i. =The ingots in question are in a state of slight meltedness. ¬@
- b. The almost unmelted ingots are the ones in the back of Storeroom C.
 - i. = Ingots that are in a state of limited meltedness ¬@
 - ii. = Ingots that were nearly missed (i.e. nearly not subjected to any melting) ¬E

Finally, we return to the contexts involving painted robots used in §5.1, repeated as (77) for convenience:

(77) 5 toy robots



²⁸Other types of modification are ungrammatical with *un*-prefixed participles. For example, recall the different types of manner adverb that can access @ and E readings; as shown in (28), both types are deviant with negated participles.

- (i) a. *This package is carefully unopened.
- b. *The secretly/recently/diligently unopened package...

This could be due to a general restriction prohibiting adverbs from modifying negated events (cf. Embick, 2004).

²⁹Another syntactic frame that sharpens what is at issue in (76a) involves *go* and a negated participle. Here, the availability of counterfactual and stative interpretations depends on the position of *almost*. This is illustrated in (i), where we have changed to $\sqrt{\text{NOTICE}}$ to maximize felicity:

- (i) Egon von Burgerhausen's third film
 - a. went almost unnoticed during his lifetime. degree: @
[=was in a state of being barely noticed during his lifetime]
 - b. almost went unnoticed during his lifetime. counterfactual: E
[=nearly was not noticed during his lifetime]

The counterfactual readings that are clear when *almost* scopes over eventive *went* are clearly not available when *almost* applies to the predicative participle by itself, which is expected given that it is not in attributive position, and hence has only a stative interpretation.

In the first version of this scenario, the only knowledge available is the visible state of the robots in (77). The attributive and predicative negated participles are then introduced, as in (78):

(78) Questions:

- a. Which robots are unpainted? (b), (d), (e) *predicative: $\neg@$*
- b. The unpainted robots are ... (b), (d), (e) *attributive: $\neg@$*

In (78), the participle in both attributive and predicative position requires an interpretation that picks out the robots that are visibly not painted: $\neg@$ ((b), (d), (e)).

We then revise the context to one in which robot (d) was painted in the recent past, with the paint having subsequently peeled off. In this updated context, the predicative participle continues to have only a negated @-reading (compare (78a) and (79a)). In contrast, the attributive participle in (79b) has an additional negated E-reading:

(79) Questions:

- a. Which robots are unpainted? (b), (d), (e) *Predicative: $\neg@$*
- b. The unpainted robots are ...
 - i. (b), (d), (e) *Attributive: $\neg@$*
 - ii. (b), (e) *Attributive: $\neg E$*

That is, while it is true that robot (d) is unpainted at Topic Time ($\neg@$ -interpretation), robot (d) was painted in the recent past, and hence cannot be said to be *unpainted* on a $\neg E$ -interpretation. These contexts confirm that distinct truth conditions arise from negating the @- and E- interpretations.

The interpretation of negated stative passives confirms the main observation advanced in the first part of this paper: attributive participles have both @- and E-interpretations, while predicative participles have only the former. In addition, negated participles provide a second case that confirms that the stative passive syntactic object can have an E- interpretation.

5.3 Summary

The evidence from agent compounding and *un*-prefixation show that (60), in its strong form, is false: at least some attributive participles that have an E-interpretation must be the stative passive participle, and cannot be the eventive passive. In the absence of compounding or negation, i.e. with simple participles like *The violently kicked ball...*, we are not aware of diagnostics for English that conclusively demonstrate that such examples must involve an eventive participle and not a stative participle, or vice versa. As far as we have been able to determine, both options appear to be possible.³⁰

6 Discussion

The primary empirical focus of this paper is on several questions raised by stative passive participles in English. The details of how these participles are interpreted connect directly with the analysis of eventive passive participles. We have argued for a syntactic analysis in which these participles share a head α , which accounts for their shared properties. At the same time, the participles differ, in ways we

³⁰Part of what is at issue stems from the fact that English stative and eventive passive participles are identical in form. This is not the case, e.g., in Modern Greek, where there are stative passive participles but no eventive passive participles. Notably, Greek shows E-interpretations for participles in attributive syntax (Paparounas 2023, submitted), further confirming (as we have argued here) that a purely eventive interpretation can be found with stative passive structures.

have reduced to the small versus phrasal distinction. One clear question for future study concerns the cross-linguistic and cross-category manifestations of this distinction. More work is needed on additional languages and further types of cross-categorical relations before general conclusions can be drawn about what is expected to be small or phrasal; some different conclusions have been reached in recent analyses, as noted in 4.

In concluding the paper we address further aspects of our analysis, in two domains. First, in §6.1, we examine the implications it produces for the theory of category; and second, in §6.2, we discuss several questions that arise concerning role of allosemy.

6.1 The theory of category

We turn now to two topics in the theory of category. The first, in §6.1.1, is general and concerns what it means to have an abstract approach to syntactic categories. The main idea is that relations between category features and other properties (argument introduction, semantic interpretation) may potentially be indirect. The second topic, addressed in §6.1.2, is more specific, and centers on the question of how our α relates to the category-defining adjectivizer a . Our goal is to outline some of the motivations for identifying these heads, and to review a number of questions that must be addressed in trying to do so.

6.1.1 Abstract categories

We have implemented an approach to category that treats the association between category labels like α and their semantic interpretations and argument introduction behavior as *indirect*. Having the head α does not necessarily entail anything in terms of semantic interpretation (stative or eventive); which is to say, structures whose distribution is determined by α are associated with different event structural interpretations. The α headed participles we have posited also differ with respect to argument introduction:

(80) properties of α (beyond distribution)

- a. semantic interpretation: α participles are interpreted as stative when i^* is present, and eventive when it is not.
- b. argument introduction: α does not introduce an argument by itself. It may or may not appear with an argument introducing head i^* (stative passive); or above structure in which an argument is introduced in a vP (eventive passive).

A certain amount of indirectness is expected on theories with abstract morphemes. By definition, abstract morphemes are purely syntactic objects that combine in syntax and receive their form and semantic interpretation at the PF and LF interfaces. So, for example, the abstract morpheme T[+past] is a single syntactic object, but it can receive more than one form and one meaning. At PF, its pronunciation may be determined by the Root that is local to it: e.g. *-t* in the context of $\sqrt{\text{LEAVE}}$ and others; *-Ø* in the context of $\sqrt{\text{HIT}}$ etc., and *-d* by default. For meaning, [+past] is often interpreted as past tense, but has irrealis nonpast interpretations as well, in e.g. conditional antecedents like *If Mary left tomorrow it would be unfortunate*.

Treating category as an abstract feature, as we have done here, produces a separation between category label and a number of other grammatical behaviors. It is perhaps easiest to explain what is at issue theoretically with reference to lexical category features like $[\pm N]$, $[\pm V]$ (cf. Chomsky (1970), and much subsequent work), which, in a certain way of thinking about lexical category, have been used to bundle together different types of effects (cf. Stowell, 1981). Taking for illustration the category Adjective, which is [+N,+V] in that system, a *direct* theory of category might hold that this feature bundle determines a rich set of properties in different parts of the grammar. For example, it might simultaneously specify (i) syntactic distribution; (ii) a specific way of introducing arguments; and (iii) a certain kind of semantic

interpretation. For the Adjective conceived of in this way, the idea is typically that the category features [+N,+V] specify first, distribution in attributive and predicative syntax; second, the licensing of external arguments; and third, that the members of this category are interpreted as states. There are other types of direct connections that can be posited beyond these; for different approaches that could be characterized as direct in the sense that we intend here, see Bešlin (2024) for a review and critique.

In the indirect approach we have argued for, each of the factors (i.-iii.) above can in principle be separated from one another. The idea that α defines a kind of syntactic distribution (which we have referred to as *non-verbal*) is a crucial part of the approach. However, what happens with α beyond this distribution is not ‘fixed’, as summarized in (80). ‘Mixed’ behaviors of these kinds are expected on an abstract approach to category, and are not expected in a direct theory of category.

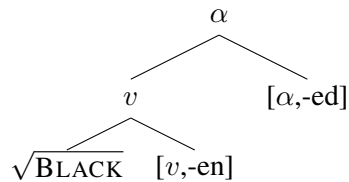
One important question—perhaps the most important question—for theories with abstract morphemes is what restrictions apply to them. With respect to this concern it is worth noting that the analysis developed here posits crucial links between the subparts of (80): argument introduction plays a crucial role in how α is interpreted, as manifested in the particular way in which we have employed contextual allomorphy. Similar links have been studied in the analysis of nominalizations. For example, it has long been noted that the availability of readings in ‘complex event nominals’ of the *examination* type is to some extent correlated with the presence or absence of arguments (see Wood, 2023, for discussion). One project for future research is to develop a general theory of category that concentrates on argument introduction, and explore different ways in which it implicates eventive and stative interpretations. Some recent forays into this domain, such as Marantz (2022), center primarily on argument introduction; it remains to be seen how theories of category of this and related types will also connect with the theory of eventualities.

6.1.2 α and the ‘adjective’

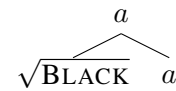
The connection between participles and adjectives is taken for granted in many grammatical traditions, and in much theoretical work as well. At the same time, there are lines of research in the syntactic literature that have pursued a verbal analysis for what we call the eventive passive participle, extending an intuition that is at play in Wasow (1977) and related work. Contrary to this idea, in this paper we have argued that both eventive and stative participles have a non-verbal distribution defined by α . More generally, it appears that there is little evidence for the verbal analysis of passive participles; see Bešlin (2023, 2024) for recent discussion.

One clear direction for further investigation concerns how participles with α like (81) relate to typical adjectives like *black*. In the approach to derivational morphology that we have assumed here, which employs Roots and categorizers, adjectives are typically treated with a head a as in (82) (and possibly functional structure above this head). How are α and a related?

(81) black-en-ed



(82) black



The most interesting answer would be to say that α and a are the same. In its simplest form, this would involve one and the same head $[\alpha]$ appearing in each. Another possibility would be to say that participles and adjectives share $[\alpha]$, but differ in terms of other features that appear with this head: using

numbers for the additional features, there might be e.g. $[\alpha, 1]$ for one, and $[\alpha, 2]$ for the other. We will return to these possibilities below, after reviewing some of the motivations for the identity hypothesis, as well as some questions that it raises.

We have seen in earlier sections of this paper why it makes sense to connect a and α in terms of syntactic distribution.³¹ We will therefore concentrate our discussion here on interface properties, as these provide an interesting set of similarities and differences to consider.

With respect to form, one point to consider are the clear overlaps in participial and certain apparently adjectival exponents. Some examples of this type appear in different places above; further examples are given in (83):

- (83) a. **Stative/eventive passive:** *paint-ed, hit-Ø, ben-t, beat-en*
 b. **Pure stative:** *bless-èd, shrunk-en, clos-ed; gold-en, silk-en, wood-en*
 c. **Denominal:** *beard-ed, three-toe-d, brown-eye-d*

Although suggestive, it is worth noting that there are also numerous exponents that appear with typical adjectives but not participles (*-ous, -al*, etc.); for that matter, the majority of adjectives do not show exponents that overlap with participial forms like those in (83). Thus, it remains to be seen precisely how to account for the formal relations that do occur, as stressed in Embick (2003).

Turning to meaning, a possible a/α relation is the fact that the @-interpretation of the stative passive is stative, and so is the interpretation of typical adjectives. The presence of the same functional head in the two structures could provide a basis for this shared property. However, a number of challenges must be addressed in making this reduction. These stem from the fact that there are many differences between the kinds of states found with participles and adjectives. One semantically-grounded distinction discussed in §2 is that only the states associated with typical adjectives, and not those defined by stative passives, can occur as resultative secondary predicates, or as depictives with verbs of creation (cf. examples (8) and (9)). The states in stative passives also tend to behave differently to simple adjectives with respect to degree structure (*very/more red (redder)* vs. *#very/#more reddened*); or *half-painted* vs. *#half-tall*.

Some further contrasts that could shed light on these matters are seen in manner modification. Recall from §3 that manner modification is possible with stative participles (84a), with the proviso that the @-reading requires compliance with State Relevance; however, manner modification is not possible at all with corresponding adjectives (84b):

- (84) a. carefully opened/expertly flattened/masterfully blackened *stative passive*
 b. *carefully open/*expertly flat/*masterfully black *pure stative*

The distribution of manner modifiers discussed in §3 indicates that it cannot be the case that such adverbs have ‘direct access’ to the event introduced by verbal structure in the complement of α . As we showed

³¹One difference involving distribution that could be investigated further concerns ordering effects. Cross-linguistically, adjectives have been observed to show some relative ordering effects with respect to (notional) lexical semantic class (*big blue* vs. *#blue big*). In contrast, stative participles do not appear display preferential ordering effects (Truswell, 2004; Cinque, 2010). Instead, stacked stative participles in prenominal position have been argued to produce temporal sequencing effects, illustrated in the asymmetric pairs in (i.) (Svenonius, 1994):

- (i) a. Chopped frozen chicken *frozen then chopped*
 b. Frozen chopped chicken *chopped then frozen*

Temporal sequencing is not associated with lexical adjectives.

One possibility to be explored is that there are two ways of interpreting sequences like those in (i): one in which there are two @-interpretations, and one in which the first participle is @, the second E. Temporal ordering would be restricted to the latter scenario.

at length, ‘direct access’ would predict that any modifier that is possible with the eventive passive should also be possible with the stative passive– in other words, that there should be no State Relevance effects– contrary to fact:

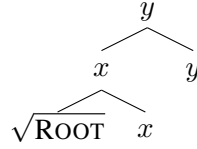
- (85) a. The X’s were secretly opened/quietly flattened/swiftly blackened. *eventive passive*
 b. #The X’s are secretly opened/quietly flattened/swiftly blackened. *stative passive*

The contrasts identified between (84a-b) on the one hand, and (85a-b) on the other, call for an analysis in which the state found in the stative passive differs from the state defined by adjectives in ways that are visible to external modifiers. The semantic interpretation that we posited for the @-state offers one way of thinking about this. We suggested that the state in the @-interpretation is the end of an event of causation, while the state of an adjective is a simple (‘uncaused’) state (see (10) in §2). END-states may evidently manifest certain properties of the event that concludes in them. This appears to be what is critically at issue in State Relevance effects. Although the details how to implement this idea are beyond the scope of the present discussion, we suggest that the facts indicate that END-states are different in type from simple states. One goal for future work is to develop this intuition formally, within the boundaries defined by the observations and proposals that we have outlined here.

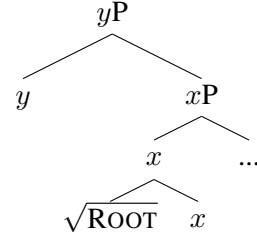
Returning to the main question posed in this section, positing the same syntactic head in participles and adjectives looks desirable for a number of reasons. The main challenge facing this thesis is that it requires re-thinking the way in which cyclicity works within the theory that we have been assuming to this point. Some background will clarify what is at issue. Embick (2003) represents the participle-forming head as Asp for both stative and eventive passives, and posits that this same Asp head occurs in ‘pure statives’ like e.g. *rotten* and *blesséd* as well. In our terms, this amounts to identifying α and a . However, this move is difficult to make on a set of assumptions that arise in more recent theories of cyclicity. In particular, many versions of Distributed Morphology have adopted a distinction between categorizing heads that are cyclic in the sense of phase theory, and other types of functional heads that are not cyclic– for review, see Embick (2021, to appear). In many versions of this kind of approach, the participial head is most straightforwardly treated as non-cyclic, because it appears outside of categorizing v , but (like, e.g. Tense), can show Root-conditioned allomorphy (cf. *beat-en*, *ben-t*, *hit-Ø*); see also Marantz (2013) for parallel considerations on the meaning side. Root-conditioned allomorphy is not predicted to occur for cyclic heads (though see below): the Root should not be able to interact with outer cyclic y . In typical adjectives, on the other hand, a standard assumption is that there is a cyclic head a present– cf. Marantz (2001), Embick and Marantz (2008). Given the apparently distinct statuses of α and a , identifying these categories calls for a careful reexamination of how the cyclic part of the theory works.

On this point, there is an interesting connection with the small versus phrasal distinction. Both Embick (2016) and Wood (2023) propose that in small objects with two categorizers x and y , when small [$\sqrt{\text{ROOT } x}$] is directly merged to y , each of these three heads should be active in the same domain; cf. (86). This corresponds to our small-stative passive, where we do indeed find Root-conditioned allomorphy. However, the same approach predicts an absence of Root-specific effects in phrasal (87), which corresponds to our eventive passive participle:

(86) small object (abstract)



(87) phrasal object (abstract)



Taken at face-value the prediction is that the eventive passive should always show a default realization of the participial head, contrary to fact.

In summary, perhaps the most important challenge to identifying α and a thus involves how to fit this with the theory of which heads are cyclic and which are not. One possibility is, of course, that adjectives and participles share a feature—our $[\alpha]$ —but differ in other features that are responsible for the cyclic/non-cyclic distinction. This and related questions can be pursued only in the context of a more detailed reworking of this part of the theory, something that goes beyond the scope of this paper. We hope at a minimum to have shown how our analysis of participles provides crucial points of reference to be taken into account in such a project.

6.2 On the role of allosemy

Our discussion of allosemy is (like the discussion of category) divided into a more general and more specific section. For the former, in §6.2.1 we situate the way in which we have employed allosemy with respect to other uses of this idea in the literature. A key question is what kinds of cross-linguistic variation (if any) should be expected in the kind of allosemy we have posited. The more specific point examined in §6.2.2 picks up on the cross-linguistic issue, and examines the kinds of predictions that might be made in theories that do and do not appeal to allosemy. In the course of this discussion, we introduce an analysis without allosemy as a direct point of comparison.

6.2.1 Context and argument introduction

Having α interpreted semantically in a way that depends on its context is a kind of allosemy. In assessing the theoretical status of this part of the analysis, it is important to take a broader perspective. The term *allosemy* has been used in different ways in the literature, and it is not clear that all of these are of the same type.

One kind of effect that has been treated as allosemy involves ‘lexical’ meaning, where morphemes local to a Root play an important role. A particularly clear instance involves ‘meaningless’ Roots like $\sqrt{\text{CEIVE}}$ (*re-ceive*, *de-ceive*, ...) or $\sqrt{\text{MIT}}$ (*com-mit*, *e-mit*,...). The same kind of point can be made with other Roots that appear to have allosemy centered on a more consistent meaning, like e.g. *stand* versus *understand* with $\sqrt{\text{STAND}}$, or e.g. $\sqrt{\text{GLOBE}}$ in *globe-Ø* vs. *glob-al* (cf. Marantz, 2013).

Another application of allosemy is found in Wood and Marantz’s (2017) approach to argument introduction. In this use, allosemy refers to the ways in which a single i^* head introduces DPs that receive different thematic interpretations, interpretations whose nature depends on the larger structure that the i^* appears in. In this second way in which allosemy has been appealed to, notably, the action is connected to the interpretation of a functional morpheme (see also Wood, 2015).

Finally, Wood’s (2023) analysis of $[[\sqrt{\text{ROOT}}\ v]\ n]$ nominalizations in Icelandic makes use of allosemy in yet another way. On this proposal, each of the category heads v or n has a semantic interpre-

tation; but each of those interpretations may be ignored (=have no semantic contribution) independently of the other. See also Benz (2023, prep) on content nominals in German.

The analysis developed in this paper is quite close to the second and third applications of allosemy. It requires that α have no interpretation when it appears by itself, and that it receive a stative interpretation only when it is the complement of i^* . While this part looks similar to analyses of allosemy in nominalizations, there are some important differences. For English participles, it is not possible in $[[\sqrt{\text{ROOT}}\ v]\ \alpha]$ to interpret α as a state and not interpret v . If this were allowed, a number of simple facts would go unexplained. For example, it predicts that *open-ed* should have an identical interpretation to *open*. There is no reason to think that this is the case, though; recall *Violeta kicked the door open/*opened*.³² Allosemy in our analysis only applies to α , and is not optional. Instead, our implementation shares properties with typical cases of contextual allomorphy, where the realization of a morpheme is produced deterministically by its structural context.

An important question for future work is how much cross-linguistic variation there is in these different applications of allosemy. For the first type, which involves contexts producing different lexical meanings, the expectation is that there will be various cross-linguistic idiosyncrasies; some lexical variation is clearly to be expected across languages (although there could of course be a universal theory of the locality conditions under which contextual meaning effects are found). For allosemy that involves interactions between functional heads, the idea that variation in the covert component of the grammar should be highly restricted (e.g. Chomsky, 1993, 1995, and much related work) provides some guidelines. All else equal, it appears that the kind of allosemy that we have appealed to should not be subject to the kind of arbitrary cross-linguistic variation seen in lexical meaning, or in allomorphy on the form side.

The strongest cross-linguistic predictions generalize the idea that small $[\alpha]$ objects are always interpreted statively when they appear with i^* , and are otherwise interpreted eventively. That is:

- (88) Strongest cross-linguistic predictions for $[\alpha]/i^*$
- a. All small $[\alpha]$ structures with i^* are interpreted statively.
 - b. All small $[\alpha]$ structures without i^* are interpreted eventively.

The possibility of testing predictions of this sort interacts with other factors. For example, if a language lacks English-type attributive syntax, the prediction in (88b) is moot. More important for our purposes, the question of cross-linguistic predictions can be asked quite generally about applications of allosemy of the kinds we have outlined; we do not believe this question has been posed often enough to have produced any sort of clear consensus, though. To sharpen what further investigation might reveal, we will now turn to an alternative to an allosemy analysis.

6.2.2 An alternative to allosemy

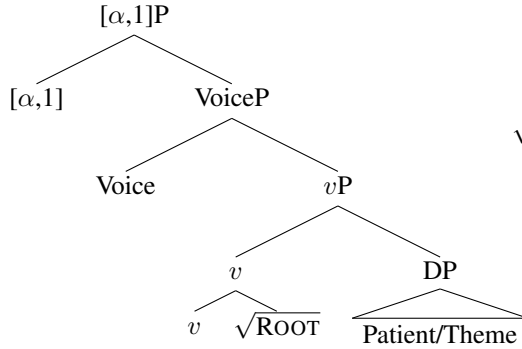
One further way to think about the cross-linguistic predictions of a theory with allosemy is to consider a syntactic analysis that does not make use of it. We will outline one such approach now; crucially, we will see that this alternative requires an indirect theory of category and makes reference to the introduction of the argument. The overall discussion thus reinforces two of our primary theoretical claims.

In the type of analysis we have in mind, the different participle interpretations are produced by featurally distinct participial heads. To implement this in a way that accounts for shared properties, it may be assumed that each of these participle-defining heads possesses the feature $[\alpha]$. To account for the

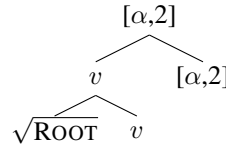
³²It is important to be careful about the form that appears in such examples. Some apparent counterexamples are found when a participle-like form is grammatical as a resultative secondary predicate, as in *Violeta kicked the door closed*. For these, it has been argued that there is syncretism between a pure stative (recall §6.1.2) and the stative passive; see Embick (2003, 2004).

differences we have identified throughout the paper, there would be additional features: using numbers [1, 2, 3] for these, the three different participle posited in this paper types would be as follows:

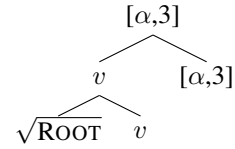
(89) Eventive passive



(90) small and stative



(91) small and eventive



With $[\alpha]$ defining a shared syntactic distribution, the additional features schematized with numbers create *subcategories* of $[\alpha]$, which produce the required differences. The work required at the interfaces for this kind of approach is as follows. First, at PF, the allomorphy patterns may be derived by underspecifying the Vocabulary Items that insert *-ed*, *-en*, etc., such that they refer only to $[\alpha]$ (cf. (49) in §4). Second, at LF the morpheme $[\alpha,2]$ is interpreted as a stativizer, while the other two participle-forming heads ($[\alpha,1]$, $[\alpha,3]$) receive no semantic interpretation.

Looking first at the prospects of (89)-(91) for the analysis of English participles, the question to examine is how the subcategory approach can connect the distinctions that are required for analysis: small vs. phrasal, stative vs. eventive, and attributive vs. predicative. For example, we argued that the E-interpretation is impossible in predicative syntax because, (i) i^* is required as an argument-introducer, there, and (ii) $[\alpha]$ is interpreted statively when i^* is present. The subcategory approach requires an alternative explanation for these correlations. Perhaps the simplest possibility is a general principle banning participles with eventive interpretations from predicative syntax; but this cannot be correct given the distribution of eventive passive participles. Taken at face value, then, it appears that the subcategory approach requires (92):

(92) $[\alpha,3]$ is not allowed in predicative syntax.

As it stands, (92) is stipulative, especially given the work that $[\alpha]$ does defining syntactic distribution more broadly. There seem to be ways of making (92) appear less stipulative, however; these start with the observations in (93):³³

(93) argument introduction (subcategory approach)

- a. $[\alpha,1]$: no i^* ; has argument introduced in the normal vP way.
- b. $[\alpha,2]$: appears as the complement of i^* , which introduces the argument.
- c. $[\alpha,3]$: Does not introduce an argument in the typical vP way or with i^*

³³In principle it would be possible to recast the subcategory approach so that differences derive from the complement of α : e.g. $[[[\sqrt{\text{ROOT}} v] 2] \alpha]$ instead of (90), where $[\alpha]$ and $[2]$ are bundled. The different interpretations would then be produced by $[1,2,3]$, with α not interpreted. On this account, though, it would not be possible to locally encode the different relationships to i^* that are stated in (93), since this head's complement would always be α .

Introducing the argument below α (cf. Embick, 2004) might alleviate this problem; but it produces phrasal structure in the complement to α , which is not possible for the reasons discussed in §3.

An analysis based on the statements in (93) differs minimally from the allosemy approach. It requires abstract categories, as forms of $[\alpha]$ may be interpreted both statively and eventively. And, like our analysis, it makes crucial reference to argument introduction in accounting for the attributive/predicative asymmetry.

With respect to the distributional asymmetry, an important question concerns precisely how the correlations in (93) are encoded. The subcategory analysis essentially accounts for the fact that predicative stative participles may not be eventive using a type of selection: $[\alpha,3]$ cannot be selected by i^* . The idea then is that $[\alpha,3]$ is banned from predicative syntax because it cannot have an argument there, unlike the other participles, which each have a way of introducing an argument in that position.

The interesting point to consider in this respect is the contrast with the strong allosemy predictions in (88). If these predictions are correct, then small stative participles are expected to never have an E-interpretation in predicative position. On the (93) type of analysis, conversely, there very well might be a language in which there is an $[\alpha,3]$ participle that is selected by i^* ; that is, a small participle in predicative position, with an eventive interpretation.

There are also some different predictions for participle forms. If there is only a single α , contextual effects for the form of α could make reference to i^* , in the way schematized in (94):

(94) possible allomorphy for α

$$\begin{aligned}\alpha &\leftrightarrow -x/__i^* \\ \alpha &\leftrightarrow -y\end{aligned}$$

That is, a language might have one form for stative passive participles with i^* – which would have stative interpretations – but another form for eventive passives and for stative passives without i^* . There are in principle more possibilities in a subcategory approach. In the limit case, each of the heads $[\alpha,1]$, $[\alpha,2]$ and $[\alpha,3]$ could have different forms; at the same time, given underspecification, the full range of syncretism possibilities could also be found.

In summary, the predictions of the allosemy and subcategory analyses differ in interesting ways, and thus provide fruitful possible lines for cross-linguistic investigation. As we noted earlier, we do not believe that the cross-linguistic picture is clear enough at present to draw general conclusions in one direction or another. Ideally the outline above will to guide investigations that speak directly to the points we have raised.

7 Conclusion

In this paper, we have developed arguments for (i) an abstract approach to syntactic categories; (ii) the role of argument introduction in accounting for how functional heads like participle-defining α are interpreted; and (iii) the small versus phrasal distinction. To connect (i) and (ii) we made use of contextual allosemy, although we have pointed out that there are alternatives. Accordingly, we have tried to clarify these different analytic possibilities and their predictions as a means of guiding further work on this topic.

On the question of further theoretical comparisons, it is notable that the alternatives considered in §6.1 and §6.2 share an interesting property. In both cases, we contrasted accounts in which a single feature $[\alpha]$ is interpreted in different ways at the interfaces with approaches in which additional features define further subcategories like $[\alpha,1]$ and so on. In its general form, the question at issue is whether to complicate the inventory of grammatical primitives, by admitting more morphemes into the building blocks of syntax; or to situate the relevant complexity at the interfaces, by allowing one and the same morpheme to be interpreted in multiple ways. In the domains considered here and much more generally,

there is no way of knowing a priori which way of doing things is correct. Instead, empirical arguments situated in the kinds of case studies provided here must prove decisive.

In the present paper, we have leaned toward the side of putting the complexity at the interfaces, given the particular set of facts that we have sought to explain. At the same time, the cross-linguistic picture is only just starting to take shape. We hope at a minimum to have shown why approaching things in the ways developed here is fruitful, and to have provided positions that are clear enough to strengthen further investigations across different languages and categories.

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